

Rectifiability of the Stratification for Harmonic Map Flows via Tangent Measures

Wei Wang

School of Mathematical Sciences,
Peking University

Joint work with Haotong Fu, Ke Wu, and Zhifei Zhang

June 8, 2025

Table of Contents

1 Background and Motivation

- Harmonic Map Flows
- Suitable Solutions
- Stratification Theory

2 Rectifiability and Minkowski Dimension

- Main Results
- Some Remarks

3 Improvements of the Regularity

- Main Results
- Some Remarks

4 Difficulties and Sketch of the Proof

- Difficulties and Strategies
- Sketch of the Proof

Harmonic Map Flows: Overview

$\Omega \subset \mathbb{R}^n$: bounded domain with $n \geq 2$.

$\mathcal{N}$: smooth, compact mfd isometrically embedded in $\mathbb{R}^d$ ($d \geq 2$).

The harmonic map flow is given by

$$\partial_t u - \Delta u = A(u)(\nabla u, \nabla u) \quad \text{in } \Omega \times \mathbb{R}_+, \quad (\text{HMF})$$

where

$$A(\cdot)(\cdot, \cdot) : \mathcal{N} \times T\mathcal{N} \times T\mathcal{N} \rightarrow (T\mathcal{N})^\perp$$

is the second fundamental form.

- **Intuition:** Evolution of maps that "minimize" the Dirichlet energy

$$\int_{\Omega} |\nabla \Phi|^2 dx, \quad \Phi \in H^1(\Omega, \mathcal{N}).$$

- **Origin:** Introduced by (Eells-Sampson, 1964) to construct harmonic maps in any given free homotopy class.
- **Physical Applications:** Connects to liquid crystal theory through the simplified Ericksen-Leslie system, coupled with the Navier-Stokes equation.

Harmonic Map Flows: Overview

$\Omega \subset \mathbb{R}^n$: bounded domain with $n \geq 2$.

$\mathcal{N}$: smooth, compact mfd isometrically embedded in $\mathbb{R}^d$ ($d \geq 2$).

The harmonic map flow is given by

$$\partial_t u - \Delta u = A(u)(\nabla u, \nabla u) \quad \text{in } \Omega \times \mathbb{R}_+, \quad (\text{HMF})$$

where

$$A(\cdot)(\cdot, \cdot) : \mathcal{N} \times T\mathcal{N} \times T\mathcal{N} \rightarrow (T\mathcal{N})^\perp$$

is the second fundamental form.

- **Intuition:** Evolution of maps that "minimize" the Dirichlet energy

$$\int_{\Omega} |\nabla \Phi|^2 dx, \quad \Phi \in H^1(\Omega, \mathcal{N}).$$

- **Origin:** Introduced by (Eells-Sampson, 1964) to construct harmonic maps in any given free homotopy class.
- **Physical Applications:** Connects to liquid crystal theory through the simplified Ericksen-Leslie system, coupled with the Navier-Stokes equation.

Harmonic Map Flows: Overview

$\Omega \subset \mathbb{R}^n$: bounded domain with $n \geq 2$.

$\mathcal{N}$: smooth, compact mfd isometrically embedded in $\mathbb{R}^d$ ($d \geq 2$).

The harmonic map flow is given by

$$\partial_t u - \Delta u = A(u)(\nabla u, \nabla u) \quad \text{in } \Omega \times \mathbb{R}_+, \quad (\text{HMF})$$

where

$$A(\cdot)(\cdot, \cdot) : \mathcal{N} \times T\mathcal{N} \times T\mathcal{N} \rightarrow (T\mathcal{N})^\perp$$

is the second fundamental form.

- **Intuition:** Evolution of maps that "minimize" the Dirichlet energy

$$\int_{\Omega} |\nabla \Phi|^2 dx, \quad \Phi \in H^1(\Omega, \mathcal{N}).$$

- **Origin:** Introduced by (Eells-Sampson, 1964) to construct harmonic maps in any given free homotopy class.
- **Physical Applications:** Connects to liquid crystal theory through the simplified Ericksen-Leslie system, coupled with the Navier-Stokes equation.

Harmonic Map Flows: Overview

$\Omega \subset \mathbb{R}^n$: bounded domain with $n \geq 2$.

$\mathcal{N}$: smooth, compact mfd isometrically embedded in $\mathbb{R}^d$ ($d \geq 2$).

The harmonic map flow is given by

$$\partial_t u - \Delta u = A(u)(\nabla u, \nabla u) \quad \text{in } \Omega \times \mathbb{R}_+, \quad (\text{HMF})$$

where

$$A(\cdot)(\cdot, \cdot) : \mathcal{N} \times T\mathcal{N} \times T\mathcal{N} \rightarrow (T\mathcal{N})^\perp$$

is the second fundamental form.

- **Intuition:** Evolution of maps that "minimize" the Dirichlet energy

$$\int_{\Omega} |\nabla \Phi|^2 dx, \quad \Phi \in H^1(\Omega, \mathcal{N}).$$

- **Origin:** Introduced by (Eells-Sampson, 1964) to construct harmonic maps in any given free homotopy class.
- **Physical Applications:** Connects to liquid crystal theory through the simplified Ericksen-Leslie system, coupled with the Navier-Stokes equation.

Key Results on Existence and Uniqueness

- (Eells-Sampson, 1964): Short-time unique smooth solutions; global smooth solutions if $K_M \leq 0$.
- (Chen-Struwe, 1989): Global weak solutions exist, smooth except on a set of the small parabolic Hausdorff measure.
- (Coron, 1990): Weak solutions may not be unique.
- (Chang-Ding-Ye, 1992): Finite-time blow-up possible, even for flows from S^2 to S^2 ! (The analysis of singularity is essential!)
- (Wang, 2008): Refined blow-up criteria.
- (Wang-Lin, 2010): Uniqueness under additional regularity conditions.

Key Results on Existence and Uniqueness

- **(Eells-Sampson, 1964):** Short-time unique smooth solutions;
global smooth solutions if $K_{\mathcal{N}} \leq 0$.
- **(Chen-Struwe, 1989):** Global weak solutions exist, smooth except on a set of the small parabolic Hausdorff measure.
- **(Coron, 1990):** Weak solutions may not be unique.
- **(Chang-Ding-Ye, 1992):** Finite-time blow-up possible, even for flows from S^2 to S^2 ! (The analysis of singularity is essential!)
- **(Wang, 2008):** Refined blow-up criteria.
- **(Wang-Lin, 2010):** Uniqueness under additional regularity conditions.

Key Results on Existence and Uniqueness

- **(Eells-Sampson, 1964)**: Short-time unique smooth solutions; global smooth solutions if $K_{\mathcal{N}} \leq 0$.
- **(Chen-Struwe, 1989)**: Global weak solutions exist, smooth except on a set of the small parabolic Hausdorff measure.
- **(Coron, 1990)**: Weak solutions may not be unique.
- **(Chang-Ding-Ye, 1992)**: Finite-time blow-up possible, even for flows from S^2 to S^2 ! (The analysis of singularity is essential!)
- **(Wang, 2008)**: Refined blow-up criteria.
- **(Wang-Lin, 2010)**: Uniqueness under additional regularity conditions.

Key Results on Existence and Uniqueness

- **(Eells-Sampson, 1964)**: Short-time unique smooth solutions; global smooth solutions if $K_{\mathcal{N}} \leq 0$.
- **(Chen-Struwe, 1989)**: Global weak solutions exist, smooth except on a set of the small parabolic Hausdorff measure.
- **(Coron, 1990)**: Weak solutions may not be unique.
- **(Chang-Ding-Ye, 1992)**: Finite-time blow-up possible, even for flows from S^2 to S^2 ! (The analysis of singularity is essential!)
- **(Wang, 2008)**: Refined blow-up criteria.
- **(Wang-Lin, 2010)**: Uniqueness under additional regularity conditions.

Key Results on Existence and Uniqueness

- **(Eells-Sampson, 1964)**: Short-time unique smooth solutions; global smooth solutions if $K_{\mathcal{N}} \leq 0$.
- **(Chen-Struwe, 1989)**: Global weak solutions exist, smooth except on a set of the small parabolic Hausdorff measure.
- **(Coron, 1990)**: Weak solutions may not be unique.
- **(Chang-Ding-Ye, 1992)**: Finite-time blow-up possible, even for flows from $\mathbb{S}^2$ to $\mathbb{S}^2$! (The analysis of singularity is essential!)
- **(Wang, 2008)**: Refined blow-up criteria.
- **(Wang-Lin, 2010)**: Uniqueness under additional regularity conditions.

Key Results on Existence and Uniqueness

- **(Eells-Sampson, 1964)**: Short-time unique smooth solutions; global smooth solutions if $K_{\mathcal{N}} \leq 0$.
- **(Chen-Struwe, 1989)**: Global weak solutions exist, smooth except on a set of the **small parabolic Hausdorff measure**.
- **(Coron, 1990)**: Weak solutions may not be unique.
- **(Chang-Ding-Ye, 1992)**: Finite-time blow-up possible, even for flows from $\mathbb{S}^2$ to $\mathbb{S}^2$! (**The analysis of singularity is essential!**)
- **(Wang, 2008)**: Refined blow-up criteria.
- **(Wang-Lin, 2010)**: Uniqueness under additional regularity conditions.

Key Results on Existence and Uniqueness

- **(Eells-Sampson, 1964)**: Short-time unique smooth solutions; global smooth solutions if $K_M \leq 0$.
- **(Chen-Struwe, 1989)**: Global weak solutions exist, smooth except on a set of the **small parabolic Hausdorff measure**.
- **(Coron, 1990)**: Weak solutions may not be unique.
- **(Chang-Ding-Ye, 1992)**: Finite-time blow-up possible, even for flows from S^2 to S^2 ! (**The analysis of singularity is essential!**)
- **(Wang, 2008)**: Refined blow-up criteria.
- **(Wang-Lin, 2010)**: Uniqueness under additional regularity conditions.

Suitable Solutions: Definition

A map $u : P_1 := B_1 \times (-1, 1) \rightarrow \mathcal{N}$ is a suitable solution of the harmonic map flow if $\partial_t u, \nabla u \in L^2(P_1)$ and:

① u is a weak solution of (HMF);

② $\forall \theta \in C_0^\infty(P_1, \mathbb{R}_{\geq 0})$,

$$\int_{P_1} (|\nabla u|^2 \partial_t \theta - 2|\partial_t u|^2 \theta - 2\partial_t u \nabla u \cdot \nabla \theta) \geq 0; \quad (\text{S1})$$

③ $\forall \xi \in C_0^\infty(P_1, \mathbb{R}^n)$,

$$\int_{P_1} (|\nabla u|^2 \operatorname{div}_x \xi - 2D\xi(\nabla u, \nabla u) + 2\partial_t u \nabla u \cdot \xi) = 0; \quad (\text{S2})$$

Remark: Introduced by (Feldman, 1994), analogous to stationary harmonic maps (Bethuel, 1993).

Suitable Solutions: Definition

A map $u : P_1 := B_1 \times (-1, 1) \rightarrow \mathcal{N}$ is a suitable solution of the harmonic map flow if $\partial_t u, \nabla u \in L^2(P_1)$ and:

① u is a weak solution of (HMF);

② $\forall \theta \in C_0^\infty(P_1, \mathbb{R}_{\geq 0}),$

$$\int_{P_1} (|\nabla u|^2 \partial_t \theta - 2|\partial_t u|^2 \theta - 2\partial_t u \nabla u \cdot \nabla \theta) \geq 0; \quad (\text{S1})$$

③ $\forall \xi \in C_0^\infty(P_1, \mathbb{R}^n),$

$$\int_{P_1} (|\nabla u|^2 \operatorname{div}_x \xi - 2D\xi(\nabla u, \nabla u) + 2\partial_t u \nabla u \cdot \xi) = 0; \quad (\text{S2})$$

Remark: Introduced by (Feldman, 1994), analogous to stationary harmonic maps (Bethuel, 1993).

Suitable Solutions: Definition

A map $u : P_1 := B_1 \times (-1, 1) \rightarrow \mathcal{N}$ is a suitable solution of the harmonic map flow if $\partial_t u, \nabla u \in L^2(P_1)$ and:

① u is a weak solution of (HMF);

② $\forall \theta \in C_0^\infty(P_1, \mathbb{R}_{\geq 0}),$

$$\int_{P_1} (|\nabla u|^2 \partial_t \theta - 2|\partial_t u|^2 \theta - 2\partial_t u \nabla u \cdot \nabla \theta) \geq 0; \quad (\text{S1})$$

③ $\forall \xi \in C_0^\infty(P_1, \mathbb{R}^n),$

$$\int_{P_1} (|\nabla u|^2 \operatorname{div}_x \xi - 2D\xi(\nabla u, \nabla u) + 2\partial_t u \nabla u \cdot \xi) = 0; \quad (\text{S2})$$

Remark: Introduced by (Feldman, 1994), analogous to stationary harmonic maps (Bethuel, 1993).

Suitable Solutions: Definition

A map $u : P_1 := B_1 \times (-1, 1) \rightarrow \mathcal{N}$ is a suitable solution of the harmonic map flow if $\partial_t u, \nabla u \in L^2(P_1)$ and:

① u is a weak solution of (HMF);

② $\forall \theta \in C_0^\infty(P_1, \mathbb{R}_{\geq 0}),$

$$\int_{P_1} (|\nabla u|^2 \partial_t \theta - 2|\partial_t u|^2 \theta - 2\partial_t u \nabla u \cdot \nabla \theta) \geq 0; \quad (\text{S1})$$

③ $\forall \xi \in C_0^\infty(P_1, \mathbb{R}^n),$

$$\int_{P_1} (|\nabla u|^2 \operatorname{div}_x \xi - 2D\xi(\nabla u, \nabla u) + 2\partial_t u \nabla u \cdot \xi) = 0; \quad (\text{S2})$$

Remark: Introduced by (Feldman, 1994), analogous to stationary harmonic maps (Bethuel, 1993).

Suitable Solutions: Definition

A map $u : P_1 := B_1 \times (-1, 1) \rightarrow \mathcal{N}$ is a suitable solution of the harmonic map flow if $\partial_t u, \nabla u \in L^2(P_1)$ and:

① u is a weak solution of (HMF);

② $\forall \theta \in C_0^\infty(P_1, \mathbb{R}_{\geq 0}),$

$$\int_{P_1} (|\nabla u|^2 \partial_t \theta - 2|\partial_t u|^2 \theta - 2\partial_t u \nabla u \cdot \nabla \theta) \geq 0; \quad (\text{S1})$$

③ $\forall \xi \in C_0^\infty(P_1, \mathbb{R}^n),$

$$\int_{P_1} (|\nabla u|^2 \operatorname{div}_x \xi - 2D\xi(\nabla u, \nabla u) + 2\partial_t u \nabla u \cdot \xi) = 0; \quad (\text{S2})$$

Remark: Introduced by (Feldman, 1994), analogous to stationary harmonic maps (Bethuel, 1993).

Properties of Suitable Solutions

Consider the perturbed map

$$u^{(\tau)}(x, t) := u(x + \tau\xi(x, t), t + \tau\theta(x, t)),$$

well-defined in P_1 for $|\tau| \ll 1$.

Define the functional corresponding to (HMF) as

$$F^{(\tau)}(u) := \int_{P_1} \left(u^{(\tau)} \partial_t u + |\nabla u^{(\tau)}|^2 \right).$$

Key properties:

- (S1) $\iff \frac{d}{d\tau} \Big|_{\tau=0} F^{(\tau)}(u) \leq 0$ when $\xi \equiv 0$.
- (S2) $\iff \frac{d}{d\tau} \Big|_{\tau=0} F^{(\tau)}(u) = 0$ when $\theta \equiv 0$.
- It follows from (S1) and (S2) that suitable solutions satisfy the **localized energy inequality** and the **monotonicity formula**.

Properties of Suitable Solutions

Consider the perturbed map

$$u^{(\tau)}(x, t) := u(x + \tau\xi(x, t), t + \tau\theta(x, t)),$$

well-defined in P_1 for $|\tau| \ll 1$.

Define the functional corresponding to (HMF) as

$$F^{(\tau)}(u) := \int_{P_1} \left(u^{(\tau)} \partial_t u + |\nabla u^{(\tau)}|^2 \right).$$

Key properties:

- (S1) $\iff \frac{d}{d\tau} \Big|_{\tau=0} F^{(\tau)}(u) \leq 0$ when $\xi \equiv 0$.
- (S2) $\iff \frac{d}{d\tau} \Big|_{\tau=0} F^{(\tau)}(u) = 0$ when $\theta \equiv 0$.
- It follows from (S1) and (S2) that suitable solutions satisfy the localized energy inequality and the monotonicity formula.

Properties of Suitable Solutions

Consider the perturbed map

$$u^{(\tau)}(x, t) := u(x + \tau\xi(x, t), t + \tau\theta(x, t)),$$

well-defined in P_1 for $|\tau| \ll 1$.

Define the functional corresponding to (HMF) as

$$F^{(\tau)}(u) := \int_{P_1} \left(u^{(\tau)} \partial_t u + |\nabla u^{(\tau)}|^2 \right).$$

Key properties:

- (S1) $\iff \frac{d}{d\tau} \Big|_{\tau=0} F^{(\tau)}(u) \leq 0$ when $\xi \equiv 0$.
- (S2) $\iff \frac{d}{d\tau} \Big|_{\tau=0} F^{(\tau)}(u) = 0$ when $\theta \equiv 0$.
- It follows from (S1) and (S2) that suitable solutions satisfy the **localized energy inequality** and the **monotonicity formula**.

Partial Regularity

Intuition: Solutions are smooth if the nonlinear term is **sufficiently small**.

For a suitable solution $u : P_{2r}(X) \rightarrow \mathcal{N}$, partial regularity is in the form as

$$r^{-n} \int_{P_{2r}(X)} |\nabla u|^2 \leq \varepsilon_0(n, \mathcal{N}) \implies u \in C^\infty(P_r(X), \mathcal{N}). \quad (\text{PR})$$

- (Feldman, 1994), (Chen-Li-Lin, 1995): (PR) holds when the target mfd is a standard sphere, i.e. $\mathcal{N} = \mathbb{S}^{d-1}$.
- (Liu, 2003): Partial regularity extends to general $\mathcal{N}$.
- **Open Problem:** Suitable solutions for any target mfd exist.
- **Singular Set:** $\mathcal{P}^n(\text{sing}(u)) = 0$, where

$$\text{sing}(u) := \{X \in P_r(X_0) : \forall \rho > 0, u \notin C^\infty(P_\rho(X))\}$$

is the singular set and $\mathcal{P}^n$ is the n -dimensional parabolic Hausdorff measure.

Partial Regularity

Intuition: Solutions are smooth if the nonlinear term is **sufficiently small**.

For a suitable solution $u : P_{2r}(X) \rightarrow \mathcal{N}$, partial regularity is in the form as

$$r^{-n} \int_{P_{2r}(X)} |\nabla u|^2 \leq \varepsilon_0(n, \mathcal{N}) \implies u \in C^\infty(P_r(X), \mathcal{N}). \quad (\text{PR})$$

- (Feldman, 1994), (Chen-Li-Lin, 1995): (PR) holds when the target mfd is a standard sphere, i.e. $\mathcal{N} = \mathbb{S}^{d-1}$.
- (Liu, 2003): Partial regularity extends to general $\mathcal{N}$.
- **Open Problem:** Suitable solutions for any target mfd exist.
- **Singular Set:** $\mathcal{P}^n(\text{sing}(u)) = 0$, where

$$\text{sing}(u) := \{X \in P_r(X_0) : \forall \rho > 0, u \notin C^\infty(P_\rho(X))\}$$

is the singular set and $\mathcal{P}^n$ is the n -dimensional parabolic Hausdorff measure.

Partial Regularity

Intuition: Solutions are smooth if the nonlinear term is **sufficiently small**.

For a suitable solution $u : P_{2r}(X) \rightarrow \mathcal{N}$, partial regularity is in the form as

$$r^{-n} \int_{P_{2r}(X)} |\nabla u|^2 \leq \varepsilon_0(n, \mathcal{N}) \implies u \in C^\infty(P_r(X), \mathcal{N}). \quad (\text{PR})$$

- **(Feldman, 1994), (Chen-Li-Lin, 1995):** (PR) holds when the target mfd is a standard sphere, i.e. $\mathcal{N} = \mathbb{S}^{d-1}$.
- **(Liu, 2003):** Partial regularity extends to general $\mathcal{N}$.
- **Open Problem:** Suitable solutions for any target mfd exist.
- **Singular Set:** $\mathcal{P}^n(\text{sing}(u)) = 0$, where

$$\text{sing}(u) := \{X \in P_r(X_0) : \forall \rho > 0, u \notin C^\infty(P_\rho(X))\}$$

is the singular set and $\mathcal{P}^n$ is the n -dimensional parabolic Hausdorff measure.

Partial Regularity

Intuition: Solutions are smooth if the nonlinear term is **sufficiently small**.

For a suitable solution $u : P_{2r}(X) \rightarrow \mathcal{N}$, partial regularity is in the form as

$$r^{-n} \int_{P_{2r}(X)} |\nabla u|^2 \leq \varepsilon_0(n, \mathcal{N}) \implies u \in C^\infty(P_r(X), \mathcal{N}). \quad (\text{PR})$$

- **(Feldman, 1994), (Chen-Li-Lin, 1995):** (PR) holds when the target mfd is a standard sphere, i.e. $\mathcal{N} = \mathbb{S}^{d-1}$.
- **(Liu, 2003):** Partial regularity extends to general $\mathcal{N}$.
- **Open Problem:** Suitable solutions for any target mfd exist.
- **Singular Set:** $\mathcal{P}^n(\text{sing}(u)) = 0$, where

$$\text{sing}(u) := \{X \in P_r(X_0) : \forall \rho > 0, u \notin C^\infty(P_\rho(X))\}$$

is the singular set and $\mathcal{P}^n$ is the n -dimensional parabolic Hausdorff measure.

Partial Regularity

Intuition: Solutions are smooth if the nonlinear term is **sufficiently small**.

For a suitable solution $u : P_{2r}(X) \rightarrow \mathcal{N}$, partial regularity is in the form as

$$r^{-n} \int_{P_{2r}(X)} |\nabla u|^2 \leq \varepsilon_0(n, \mathcal{N}) \implies u \in C^\infty(P_r(X), \mathcal{N}). \quad (\text{PR})$$

- **(Feldman, 1994), (Chen-Li-Lin, 1995):** (PR) holds when the target mfd is a standard sphere, i.e. $\mathcal{N} = \mathbb{S}^{d-1}$.
- **(Liu, 2003):** Partial regularity extends to general $\mathcal{N}$.
- **Open Problem:** Suitable solutions for any target mfds exist.
- **Singular Set:** $\mathcal{P}^n(\text{sing}(u)) = 0$, where

$$\text{sing}(u) := \{X \in P_r(X_0) : \forall \rho > 0, u \notin C^\infty(P_\rho(X))\}$$

is the singular set and $\mathcal{P}^n$ is the n -dimensional parabolic Hausdorff measure.

Partial Regularity

Intuition: Solutions are smooth if the nonlinear term is **sufficiently small**.

For a suitable solution $u : P_{2r}(X) \rightarrow \mathcal{N}$, partial regularity is in the form as

$$r^{-n} \int_{P_{2r}(X)} |\nabla u|^2 \leq \varepsilon_0(n, \mathcal{N}) \implies u \in C^\infty(P_r(X), \mathcal{N}). \quad (\text{PR})$$

- **(Feldman, 1994), (Chen-Li-Lin, 1995):** (PR) holds when the target mfd is a standard sphere, i.e. $\mathcal{N} = \mathbb{S}^{d-1}$.
- **(Liu, 2003):** Partial regularity extends to general $\mathcal{N}$.
- **Open Problem:** Suitable solutions for any target mfd exist.
- **Singular Set:** $\mathcal{P}^n(\text{sing}(u)) = 0$, where

$$\text{sing}(u) := \{X \in P_r(X_0) : \forall \rho > 0, u \notin C^\infty(P_\rho(X))\}$$

is the singular set and $\mathcal{P}^n$ is the n -dimensional parabolic Hausdorff measure.

Blow-up Analysis

$U \subset \mathbb{R}^k$: open set. $\mathcal{M}(U)$: the space of Radon measures on U .

For $X_0 = (x_0, t_0) \in \mathbb{R}^n \times \mathbb{R}$ and $r > 0$, let

$$\mathcal{P}_{X_0, r}(X) = (x_0 + rx, t_0 + r^2 t), \quad X = (x, t) \in \mathbb{R}^n \times \mathbb{R}.$$

For a suitable solution $u : P_1 \rightarrow \mathcal{N}$, $\mu \in \mathcal{M}(P_1)$, $r > 0$, and $X \in P_1$, define

$$T_{X, r} u := u \circ \mathcal{P}_{X_0, r}, \quad T_{X, r} \mu := r^{-n} (\mu \circ \mathcal{P}_{X_0, r}).$$

Then, $\exists r_i \rightarrow 0^+$ s.t.

- $w_i := T_{X, r_i} u \rightharpoonup w$ weakly in $H_{\text{loc}}^1(\mathbb{R}^n \times \mathbb{R}, \mathbb{R}^d)$.
- $\frac{1}{2} |\nabla w_i|^2 dx dt \rightharpoonup^* \eta := \frac{1}{2} |\nabla w|^2 dx dt + \nu$ in $\mathcal{M}(\mathbb{R}^n \times \mathbb{R})$.
- w : tangent flow (TF) of u at X .
- η : tangent measure (TM) of u at X .
- ν : defect measure.

Blow-up Analysis

$U \subset \mathbb{R}^k$: open set. $\mathcal{M}(U)$: the space of Radon measures on U .

For $X_0 = (x_0, t_0) \in \mathbb{R}^n \times \mathbb{R}$ and $r > 0$, let

$$\mathcal{P}_{X_0, r}(X) = (x_0 + rx, t_0 + r^2 t), \quad X = (x, t) \in \mathbb{R}^n \times \mathbb{R}.$$

For a suitable solution $u : P_1 \rightarrow \mathcal{N}$, $\mu \in \mathcal{M}(P_1)$, $r > 0$, and $X \in P_1$, define

$$T_{X, r} u := u \circ \mathcal{P}_{X_0, r}, \quad T_{X, r} \mu := r^{-n} (\mu \circ \mathcal{P}_{X_0, r}).$$

Then, $\exists r_i \rightarrow 0^+$ s.t.

- $w_i := T_{X, r_i} u \rightharpoonup w$ weakly in $H_{\text{loc}}^1(\mathbb{R}^n \times \mathbb{R}, \mathbb{R}^d)$.
- $\frac{1}{2} |\nabla w_i|^2 dx dt \rightharpoonup^* \eta := \frac{1}{2} |\nabla w|^2 dx dt + \nu$ in $\mathcal{M}(\mathbb{R}^n \times \mathbb{R})$.
- w : tangent flow (TF) of u at X .
- η : tangent measure (TM) of u at X .
- ν : defect measure.

Blow-up Analysis

$U \subset \mathbb{R}^k$: open set. $\mathcal{M}(U)$: the space of Radon measures on U .

For $X_0 = (x_0, t_0) \in \mathbb{R}^n \times \mathbb{R}$ and $r > 0$, let

$$\mathcal{P}_{X_0, r}(X) = (x_0 + rx, t_0 + r^2 t), \quad X = (x, t) \in \mathbb{R}^n \times \mathbb{R}.$$

For a suitable solution $u : P_1 \rightarrow \mathcal{N}$, $\mu \in \mathcal{M}(P_1)$, $r > 0$, and $X \in P_1$, define

$$T_{X, r} u := u \circ \mathcal{P}_{X_0, r}, \quad T_{X, r} \mu := r^{-n} (\mu \circ \mathcal{P}_{X_0, r}).$$

Then, $\exists r_i \rightarrow 0^+$ s.t.

- $w_i := T_{X, r_i} u \rightharpoonup w$ weakly in $H_{\text{loc}}^1(\mathbb{R}^n \times \mathbb{R}, \mathbb{R}^d)$.
- $\frac{1}{2} |\nabla w_i|^2 dx dt \rightharpoonup^* \eta := \frac{1}{2} |\nabla w|^2 dx dt + \nu$ in $\mathcal{M}(\mathbb{R}^n \times \mathbb{R})$.
- w : tangent flow (TF) of u at X .
- η : tangent measure (TM) of u at X .
- ν : defect measure.

Blow-up Analysis

$U \subset \mathbb{R}^k$: open set. $\mathcal{M}(U)$: the space of Radon measures on U .

For $X_0 = (x_0, t_0) \in \mathbb{R}^n \times \mathbb{R}$ and $r > 0$, let

$$\mathcal{P}_{X_0, r}(X) = (x_0 + rx, t_0 + r^2 t), \quad X = (x, t) \in \mathbb{R}^n \times \mathbb{R}.$$

For a suitable solution $u : P_1 \rightarrow \mathcal{N}$, $\mu \in \mathcal{M}(P_1)$, $r > 0$, and $X \in P_1$, define

$$T_{X, r} u := u \circ \mathcal{P}_{X_0, r}, \quad T_{X, r} \mu := r^{-n} (\mu \circ \mathcal{P}_{X_0, r}).$$

Then, $\exists r_i \rightarrow 0^+$ s.t.

- $w_i := T_{X, r_i} u \rightharpoonup w$ weakly in $H_{\text{loc}}^1(\mathbb{R}^n \times \mathbb{R}, \mathbb{R}^d)$.
- $\frac{1}{2} |\nabla w_i|^2 dx dt \rightharpoonup^* \eta := \frac{1}{2} |\nabla w|^2 dx dt + \nu$ in $\mathcal{M}(\mathbb{R}^n \times \mathbb{R})$.
- w : tangent flow (TF) of u at X .
- η : tangent measure (TM) of u at X .
- ν : defect measure.

Symmetry Definitions

- **Backwardly Self-Similar:** A map $w : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathcal{N}$ satisfies that $\forall \lambda > 0$, $T_{X,\lambda} w = T_{X,1} w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similarly for $\mu \in \mathcal{M}(\mathbb{R}^n \times \mathbb{R})$.
- **Backward Invariance:** Let $V \subset \mathbb{R}^n$ be a k -dimensional subspace. $\forall v \in V$, $w(\cdot + v, \cdot) = w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similar for measures.
- **$V(w)$:** Maximal subspace of the backward invariance. Similar for measures.
- **Spatial k -Symmetry:** w (or μ) is backwardly self-similar with $\dim(V(w)) \geq k$ (or $\dim(V(\mu)) \geq k$).
- **Static, Quasi-Static, Shrinking:** w (or μ) is static if time-independent, quasi-static if time-independent up to some T , or shrinking otherwise.
- **Space-Time k -Symmetry:** $\dim(V(w)) + 2$ (static) or $\dim(V(w))$ (non-static) $\geq k$ (in the static case, an additional symmetry property exists in the time variable). Similar for measures.

Symmetry Definitions

- **Backwardly Self-Similar:** A map $w : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathcal{N}$ satisfies that $\forall \lambda > 0$, $T_{X,\lambda} w = T_{X,1} w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similarly for $\mu \in \mathcal{M}(\mathbb{R}^n \times \mathbb{R})$.
- **Backward Invariance:** Let $V \subset \mathbb{R}^n$ be a k -dimensional subspace. $\forall v \in V$, $w(\cdot + v, \cdot) = w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similar for measures.
- **$V(w)$:** Maximal subspace of the backward invariance. Similar for measures.
- **Spatial k -Symmetry:** w (or μ) is backwardly self-similar with $\dim(V(w)) \geq k$ (or $\dim(V(\mu)) \geq k$).
- **Static, Quasi-Static, Shrinking:** w (or μ) is static if time-independent, quasi-static if time-independent up to some T , or shrinking otherwise.
- **Space-Time k -Symmetry:** $\dim(V(w)) + 2$ (static) or $\dim(V(w))$ (non-static) $\geq k$ (in the static case, an additional symmetry property exists in the time variable). Similar for measures.

Symmetry Definitions

- **Backwardly Self-Similar:** A map $w : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathcal{N}$ satisfies that $\forall \lambda > 0$, $T_{X,\lambda} w = T_{X,1} w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similarly for $\mu \in \mathcal{M}(\mathbb{R}^n \times \mathbb{R})$.
- **Backward Invariance:** Let $V \subset \mathbb{R}^n$ be a k -dimensional subspace. $\forall v \in V$, $w(\cdot + v, \cdot) = w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similar for measures.
- **$V(w)$:** Maximal subspace of the backward invariance. Similar for measures.
- **Spatial k -Symmetry:** w (or μ) is backwardly self-similar with $\dim(V(w)) \geq k$ (or $\dim(V(\mu)) \geq k$).
- **Static, Quasi-Static, Shrinking:** w (or μ) is static if time-independent, quasi-static if time-independent up to some T , or shrinking otherwise.
- **Space-Time k -Symmetry:** $\dim(V(w)) + 2$ (static) or $\dim(V(w))$ (non-static) $\geq k$ (in the static case, an additional symmetry property exists in the time variable). Similar for measures.

Symmetry Definitions

- **Backwardly Self-Similar:** A map $w : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathcal{N}$ satisfies that $\forall \lambda > 0$, $T_{X,\lambda} w = T_{X,1} w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similarly for $\mu \in \mathcal{M}(\mathbb{R}^n \times \mathbb{R})$.
- **Backward Invariance:** Let $V \subset \mathbb{R}^n$ be a k -dimensional subspace. $\forall v \in V$, $w(\cdot + v, \cdot) = w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similar for measures.
- **$V(w)$:** Maximal subspace of the backward invariance. Similar for measures.
- **Spatial k -Symmetry:** w (or μ) is backwardly self-similar with $\dim(V(w)) \geq k$ (or $\dim(V(\mu)) \geq k$).
- **Static, Quasi-Static, Shrinking:** w (or μ) is static if time-independent, quasi-static if time-independent up to some T , or shrinking otherwise.
- **Space-Time k -Symmetry:** $\dim(V(w)) + 2$ (static) or $\dim(V(w))$ (non-static) $\geq k$ (in the static case, an additional symmetry property exists in the time variable). Similar for measures.

Symmetry Definitions

- **Backwardly Self-Similar:** A map $w : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathcal{N}$ satisfies that $\forall \lambda > 0$, $T_{X,\lambda} w = T_{X,1} w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similarly for $\mu \in \mathcal{M}(\mathbb{R}^n \times \mathbb{R})$.
- **Backward Invariance:** Let $V \subset \mathbb{R}^n$ be a k -dimensional subspace. $\forall v \in V$, $w(\cdot + v, \cdot) = w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similar for measures.
- **$V(w)$:** Maximal subspace of the backward invariance. Similar for measures.
- **Spatial k -Symmetry:** w (or μ) is backwardly self-similar with $\dim(V(w)) \geq k$ (or $\dim(V(\mu)) \geq k$).
- **Static, Quasi-Static, Shrinking:** w (or μ) is static if time-independent, quasi-static if time-independent up to some T , or shrinking otherwise.
- **Space-Time k -Symmetry:** $\dim(V(w)) + 2$ (static) or $\dim(V(w))$ (non-static) $\geq k$ (in the static case, an additional symmetry property exists in the time variable). Similar for measures.

Symmetry Definitions

- **Backwardly Self-Similar:** A map $w : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathcal{N}$ satisfies that $\forall \lambda > 0$, $T_{X,\lambda} w = T_{X,1} w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similarly for $\mu \in \mathcal{M}(\mathbb{R}^n \times \mathbb{R})$.
- **Backward Invariance:** Let $V \subset \mathbb{R}^n$ be a k -dimensional subspace. $\forall v \in V$, $w(\cdot + v, \cdot) = w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similar for measures.
- **$V(w)$:** Maximal subspace of the backward invariance. Similar for measures.
- **Spatial k -Symmetry:** w (or μ) is backwardly self-similar with $\dim(V(w)) \geq k$ (or $\dim(V(\mu)) \geq k$).
- **Static, Quasi-Static, Shrinking:** w (or μ) is static if time-independent, quasi-static if time-independent up to some T , or shrinking otherwise.
- **Space-Time k -Symmetry:** $\dim(V(w)) + 2$ (static) or $\dim(V(w))$ (non-static) $\geq k$ (in the static case, an additional symmetry property exists in the time variable). Similar for measures.

Symmetry Definitions

- **Backwardly Self-Similar:** A map $w : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathcal{N}$ satisfies that $\forall \lambda > 0$, $T_{X,\lambda} w = T_{X,1} w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similarly for $\mu \in \mathcal{M}(\mathbb{R}^n \times \mathbb{R})$.
- **Backward Invariance:** Let $V \subset \mathbb{R}^n$ be a k -dimensional subspace. $\forall v \in V$, $w(\cdot + v, \cdot) = w$ in $\mathbb{R}^n \times \mathbb{R}_-$. Similar for measures.
- **$V(w)$:** Maximal subspace of the backward invariance. Similar for measures.
- **Spatial k -Symmetry:** w (or μ) is backwardly self-similar with $\dim(V(w)) \geq k$ (or $\dim(V(\mu)) \geq k$).
- **Static, Quasi-Static, Shrinking:** w (or μ) is static if time-independent, quasi-static if time-independent up to some T , or shrinking otherwise.
- **Space-Time k -Symmetry:** $\dim(V(w)) + 2$ (static) or $\dim(V(w))$ (non-static) $\geq k$ (in the static case, an additional symmetry property exists in the time variable). Similar for measures.

Stratification: Definitions

$$k \in \mathbb{Z} \cap [0, n+2]$$

$u : P_1 \rightarrow \mathcal{N}$: suitable solution of (HMF).

- **Space-Time Strata:**

$$S^k(u) := \{X \in P_1 : \text{no TF at } X \text{ is space-time } (k+1)\text{-sym}\},$$

$$\Sigma^k(u) := \{X \in P_1 : \text{no TM at } X \text{ is space-time } (k+1)\text{-sym}\}.$$

- **Spatial Strata** (for $t \in (-1, 1)$):

$$S^k(u, t) := \{x \in B_1 : \text{no TF at } (x, t) \text{ is spatial } (k+1)\text{-sym}\},$$

$$\Sigma^k(u, t) := \{x \in B_1 : \text{no TM at } (x, t) \text{ is spatial } (k+1)\text{-sym}\}.$$

- **Inclusion Property of S^k and Σ^k :**

$$S^0(u) \subset S^1(u) \subset \cdots \subset S^{n+1}(u) \subset \text{sing}(u),$$

$$\Sigma^0(u) \subset \Sigma^1(u) \subset \cdots \subset \Sigma^n(u) = \text{sing}(u).$$

Note: Vanishing TF $\nRightarrow$ regular point!

Stratification: Definitions

$$k \in \mathbb{Z} \cap [0, n+2]$$

$u : P_1 \rightarrow \mathcal{N}$: suitable solution of (HMF).

- **Space-Time Strata:**

$$S^k(u) := \{X \in P_1 : \text{no TF at } X \text{ is space-time } (k+1)\text{-sym}\},$$

$$\Sigma^k(u) := \{X \in P_1 : \text{no TM at } X \text{ is space-time } (k+1)\text{-sym}\}.$$

- **Spatial Strata** (for $t \in (-1, 1)$):

$$S^k(u, t) := \{x \in B_1 : \text{no TF at } (x, t) \text{ is spatial } (k+1)\text{-sym}\},$$

$$\Sigma^k(u, t) := \{x \in B_1 : \text{no TM at } (x, t) \text{ is spatial } (k+1)\text{-sym}\}.$$

- **Inclusion Property of S^k and Σ^k :**

$$S^0(u) \subset S^1(u) \subset \cdots \subset S^{n+1}(u) \subset \text{sing}(u),$$

$$\Sigma^0(u) \subset \Sigma^1(u) \subset \cdots \subset \Sigma^n(u) = \text{sing}(u).$$

Note: Vanishing TF $\nRightarrow$ regular point!

Stratification: Definitions

$$k \in \mathbb{Z} \cap [0, n+2]$$

$u : P_1 \rightarrow \mathcal{N}$: suitable solution of (HMF).

- **Space-Time Strata:**

$$S^k(u) := \{X \in P_1 : \text{no TF at } X \text{ is space-time } (k+1)\text{-sym}\},$$

$$\Sigma^k(u) := \{X \in P_1 : \text{no TM at } X \text{ is space-time } (k+1)\text{-sym}\}.$$

- **Spatial Strata** (for $t \in (-1, 1)$):

$$S^k(u, t) := \{x \in B_1 : \text{no TF at } (x, t) \text{ is spatial } (k+1)\text{-sym}\},$$

$$\Sigma^k(u, t) := \{x \in B_1 : \text{no TM at } (x, t) \text{ is spatial } (k+1)\text{-sym}\}.$$

- **Inclusion Property of S^k and Σ^k :**

$$S^0(u) \subset S^1(u) \subset \cdots \subset S^{n+1}(u) \subset \text{sing}(u),$$

$$\Sigma^0(u) \subset \Sigma^1(u) \subset \cdots \subset \Sigma^n(u) = \text{sing}(u).$$

Note: Vanishing TF $\nRightarrow$ regular point!

Stratification: Definitions

$$k \in \mathbb{Z} \cap [0, n+2]$$

$u : P_1 \rightarrow \mathcal{N}$: suitable solution of (HMF).

- **Space-Time Strata:**

$$S^k(u) := \{X \in P_1 : \text{no TF at } X \text{ is space-time } (k+1)\text{-sym}\},$$

$$\Sigma^k(u) := \{X \in P_1 : \text{no TM at } X \text{ is space-time } (k+1)\text{-sym}\}.$$

- **Spatial Strata** (for $t \in (-1, 1)$):

$$S^k(u, t) := \{x \in B_1 : \text{no TF at } (x, t) \text{ is spatial } (k+1)\text{-sym}\},$$

$$\Sigma^k(u, t) := \{x \in B_1 : \text{no TM at } (x, t) \text{ is spatial } (k+1)\text{-sym}\}.$$

- **Inclusion Property of S^k and Σ^k :**

$$S^0(u) \subset S^1(u) \subset \cdots \subset S^{n+1}(u) \subset \text{sing}(u),$$

$$\Sigma^0(u) \subset \Sigma^1(u) \subset \cdots \subset \Sigma^n(u) = \text{sing}(u).$$

Note: Vanishing TF $\nRightarrow$ regular point!

Dimension Estimates

$\dim_{\mathcal{H}}$: Hausdorff dimension.

$\dim_{\mathcal{P}}$: Parabolic Hausdorff dimension.

$\dim_{\text{Min}}$: Minkowski dimension.

$\dim_{\text{PMin}}$: Parabolic Minkowski dimension.

Using Federer dimension reduction:

- $\dim_{\mathcal{P}} S^k(u), \Sigma^k(u) \leq k$;
- $\dim_{\mathcal{H}} S^k(u, t), \Sigma^k(u, t) \leq k \ \forall t$;
- $\dim_{\mathcal{H}} S^k(u) \cap \{t\}, \Sigma^k(u) \cap \{t\} \leq k - 2$ for a.e. t .

Question

Can we further characterize the strata and singular sets of suitable solutions?

- $\mathcal{P}^n(\text{sing}(u)) = 0 \implies \mathcal{H}^{n-2}(\text{sing}(u) \cap \{t\}) = 0$ for a.e. t .
- **(Cheeger-Haslhofer-Naber, 2015):**
 $\dim_{\text{PMin}} S^n(u) \leq n$ via quantitative stratification method.

Dimension Estimates

$\dim_{\mathcal{H}}$: Hausdorff dimension.

$\dim_{\mathcal{P}}$: Parabolic Hausdorff dimension.

$\dim_{\text{Min}}$: Minkowski dimension.

$\dim_{\text{PMin}}$: Parabolic Minkowski dimension.

Using **Federer dimension reduction**:

- $\dim_{\mathcal{P}} S^k(u), \Sigma^k(u) \leq k$;
- $\dim_{\mathcal{H}} S^k(u, t), \Sigma^k(u, t) \leq k \ \forall t$;
- $\dim_{\mathcal{H}} S^k(u) \cap \{t\}, \Sigma^k(u) \cap \{t\} \leq k - 2$ for a.e. t .

Question

Can we further characterize the strata and singular sets of suitable solutions?

- $\mathcal{P}^n(\text{sing}(u)) = 0 \implies \mathcal{H}^{n-2}(\text{sing}(u) \cap \{t\}) = 0$ for a.e. t .
- **(Cheeger-Haslhofer-Naber, 2015):**
 $\dim_{\text{PMin}} S^n(u) \leq n$ via **quantitative stratification method**.

Dimension Estimates

$\dim_{\mathcal{H}}$: Hausdorff dimension.

$\dim_{\mathcal{P}}$: Parabolic Hausdorff dimension.

$\dim_{\text{Min}}$: Minkowski dimension.

$\dim_{\text{PMin}}$: Parabolic Minkowski dimension.

Using **Federer dimension reduction**:

- $\dim_{\mathcal{P}} S^k(u), \Sigma^k(u) \leq k$;
- $\dim_{\mathcal{H}} S^k(u, t), \Sigma^k(u, t) \leq k \ \forall t$;
- $\dim_{\mathcal{H}} S^k(u) \cap \{t\}, \Sigma^k(u) \cap \{t\} \leq k - 2$ for a.e. t .

Question

Can we further characterize the strata and singular sets of suitable solutions?

- $\mathcal{P}^n(\text{sing}(u)) = 0 \implies \mathcal{H}^{n-2}(\text{sing}(u) \cap \{t\}) = 0$ for a.e. t .
- **(Cheeger-Haslhofer-Naber, 2015):**
 $\dim_{\text{PMin}} S^n(u) \leq n$ via **quantitative stratification method**.

Dimension Estimates

$\dim_{\mathcal{H}}$: Hausdorff dimension.

$\dim_{\mathcal{P}}$: Parabolic Hausdorff dimension.

$\dim_{\text{Min}}$: Minkowski dimension.

$\dim_{\text{PMin}}$: Parabolic Minkowski dimension.

Using **Federer dimension reduction**:

- $\dim_{\mathcal{P}} S^k(u), \Sigma^k(u) \leq k$;
- $\dim_{\mathcal{H}} S^k(u, t), \Sigma^k(u, t) \leq k \ \forall t$;
- $\dim_{\mathcal{H}} S^k(u) \cap \{t\}, \Sigma^k(u) \cap \{t\} \leq k - 2$ for a.e. t .

Question

Can we further characterize the strata and singular sets of suitable solutions?

- $\mathcal{P}^n(\text{sing}(u)) = 0 \implies \mathcal{H}^{n-2}(\text{sing}(u) \cap \{t\}) = 0$ for a.e. t .
- **(Cheeger-Haslhofer-Naber, 2015):**
 $\dim_{\text{PMin}} S^n(u) \leq n$ via **quantitative stratification method**.

Table of Contents

1 Background and Motivation

- Harmonic Map Flows
- Suitable Solutions
- Stratification Theory

2 Rectifiability and Minkowski Dimension

- Main Results
- Some Remarks

3 Improvements of the Regularity

- Main Results
- Some Remarks

4 Difficulties and Sketch of the Proof

- Difficulties and Strategies
- Sketch of the Proof

Rectifiability: Definition

A set $M \subset \mathbb{R}^N$ is k -rectifiable if

$$M \subset M_0 \cup \bigcup_{i \in \mathbb{Z}_+} f_i(\mathbb{R}^k),$$

where $\mathcal{H}^k(M_0) = 0$ and $f_i : \mathbb{R}^k \rightarrow \mathbb{R}^N$ are Lipschitz maps.

- **Note:** Rectifiable sets behave like manifolds for analytical purposes.
- **Remark:** Lipschitz can be replaced by C^1 .

Main Theorem: Rectifiability and Minkowski Dimension

Theorem (Fu-W.-Wu-Zhang, arXiv:2504.14880)

$u : P_1 \rightarrow \mathcal{N}$: *suitable solution* of (HMF).

$\text{Min}_\rho^k(\cdot)$: Minkowski content.

$t \in (-1, 1)$, $\Lambda > 0$.

① *Inclusion*: $\Sigma^0(u, t) \times \{t\} \subset \cdots \subset \Sigma^{n-2}(u, t) \times \{t\} = \text{sing}(u) \cap \{t\}$.

② *Rectifiability*:

- $\Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$;
- $\text{sing}(u) \cap \{t\}$ is $(n-2)$ -rectifiable.

③ *Minkowski Dimension*: If $\|\nabla u\|_{L^2(P_1)} \leq \Lambda$, then

$$\begin{aligned} \text{Min}_r^{n-2}(\text{sing}(u) \cap (B_1 \times \{t\})) &\leq C(\Lambda, n, \mathcal{N}), \\ \implies \dim_{\text{Min}}(\text{sing}(u) \cap \{t\}) &\leq n-2. \end{aligned}$$

Main Theorem: Rectifiability and Minkowski Dimension

Theorem (Fu-W.-Wu-Zhang, arXiv:2504.14880)

$u : P_1 \rightarrow \mathcal{N}$: *suitable solution* of (HMF).

$\text{Min}_\rho^k(\cdot)$: Minkowski content.

$t \in (-1, 1)$, $\Lambda > 0$.

① **Inclusion:** $\Sigma^0(u, t) \times \{t\} \subset \cdots \subset \Sigma^{n-2}(u, t) \times \{t\} = \text{sing}(u) \cap \{t\}$.

② **Rectifiability:**

- $\Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$;
- $\text{sing}(u) \cap \{t\}$ is $(n-2)$ -rectifiable.

③ **Minkowski Dimension:** If $\|\nabla u\|_{L^2(P_1)} \leq \Lambda$, then

$$\begin{aligned} \text{Min}_r^{n-2}(\text{sing}(u) \cap (B_1 \times \{t\})) &\leq C(\Lambda, n, \mathcal{N}), \\ \implies \dim_{\text{Min}}(\text{sing}(u) \cap \{t\}) &\leq n-2. \end{aligned}$$

Main Theorem: Rectifiability and Minkowski Dimension

Theorem (Fu-W.-Wu-Zhang, arXiv:2504.14880)

$u : P_1 \rightarrow \mathcal{N}$: *suitable solution* of (HMF).

$\text{Min}_\rho^k(\cdot)$: Minkowski content.

$t \in (-1, 1)$, $\Lambda > 0$.

① **Inclusion:** $\Sigma^0(u, t) \times \{t\} \subset \cdots \subset \Sigma^{n-2}(u, t) \times \{t\} = \text{sing}(u) \cap \{t\}$.

② **Rectifiability:**

- $\Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$;
- $\text{sing}(u) \cap \{t\}$ is $(n-2)$ -rectifiable.

③ **Minkowski Dimension:** If $\|\nabla u\|_{L^2(P_1)} \leq \Lambda$, then

$$\begin{aligned} \text{Min}_r^{n-2}(\text{sing}(u) \cap (B_1 \times \{t\})) &\leq C(\Lambda, n, \mathcal{N}), \\ \implies \dim_{\text{Min}}(\text{sing}(u) \cap \{t\}) &\leq n-2. \end{aligned}$$

Main Theorem: Rectifiability and Minkowski Dimension

Theorem (Fu-W.-Wu-Zhang, arXiv:2504.14880)

$u : P_1 \rightarrow \mathcal{N}$: *suitable solution* of (HMF).

$\text{Min}_\rho^k(\cdot)$: Minkowski content.

$t \in (-1, 1)$, $\Lambda > 0$.

① **Inclusion:** $\Sigma^0(u, t) \times \{t\} \subset \cdots \subset \Sigma^{n-2}(u, t) \times \{t\} = \text{sing}(u) \cap \{t\}$.

② **Rectifiability:**

- $\Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$;
- $\text{sing}(u) \cap \{t\}$ is $(n-2)$ -rectifiable.

③ **Minkowski Dimension:** If $\|\nabla u\|_{L^2(P_1)} \leq \Lambda$, then

$$\begin{aligned} \text{Min}_r^{n-2}(\text{sing}(u) \cap (B_1 \times \{t\})) &\leq C(\Lambda, n, \mathcal{N}), \\ \implies \dim_{\text{Min}}(\text{sing}(u) \cap \{t\}) &\leq n-2. \end{aligned}$$

Main Theorem: Rectifiability and Minkowski Dimension

Theorem (Fu-W.-Wu-Zhang, arXiv:2504.14880)

$u : P_1 \rightarrow \mathcal{N}$: *suitable solution* of (HMF).

$\text{Min}_\rho^k(\cdot)$: Minkowski content.

$t \in (-1, 1)$, $\Lambda > 0$.

① **Inclusion:** $\Sigma^0(u, t) \times \{t\} \subset \cdots \subset \Sigma^{n-2}(u, t) \times \{t\} = \text{sing}(u) \cap \{t\}$.

② **Rectifiability:**

- $\Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$;
- $\text{sing}(u) \cap \{t\}$ is $(n-2)$ -rectifiable.

③ **Minkowski Dimension:** If $\|\nabla u\|_{L^2(P_1)} \leq \Lambda$, then

$$\begin{aligned} \text{Min}_r^{n-2}(\text{sing}(u) \cap (B_1 \times \{t\})) &\leq C(\Lambda, n, \mathcal{N}), \\ \implies \dim_{\text{Min}}(\text{sing}(u) \cap \{t\}) &\leq n-2. \end{aligned}$$

Main Theorem: Rectifiability and Minkowski Dimension

Theorem (Fu-W.-Wu-Zhang, arXiv:2504.14880)

$u : P_1 \rightarrow \mathcal{N}$: *suitable solution* of (HMF).

$\text{Min}_\rho^k(\cdot)$: Minkowski content.

$t \in (-1, 1)$, $\Lambda > 0$.

① **Inclusion:** $\Sigma^0(u, t) \times \{t\} \subset \cdots \subset \Sigma^{n-2}(u, t) \times \{t\} = \text{sing}(u) \cap \{t\}$.

② **Rectifiability:**

- $\Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$;
- $\text{sing}(u) \cap \{t\}$ is $(n-2)$ -rectifiable.

③ **Minkowski Dimension:** If $\|\nabla u\|_{L^2(P_1)} \leq \Lambda$, then

$$\begin{aligned} \text{Min}_r^{n-2}(\text{sing}(u) \cap (B_1 \times \{t\})) &\leq C(\Lambda, n, \mathcal{N}), \\ \implies \dim_{\text{Min}}(\text{sing}(u) \cap \{t\}) &\leq n-2. \end{aligned}$$

Remarks on Stratification

- **Top Stratum:** $\Sigma^{n-2}(u, t)$ is the top spatial stratum, implying that tangent measures with spatial $(n-1)$ -symmetry vanish.
- **Sharpness:** Since $\dim_{\mathcal{H}} \Sigma^k(u, t) \leq k$, the rectifiability and the Minkowski dimension bound are all **sharp**.
- **Via TF:** $S^k(u, t) \subset \Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$.
- **Uniqueness of TF:** We do not require the uniqueness of TF.
- **Harmonic Maps:** Assuming $\partial_t u \equiv 0$, the properties degenerate to those in (Naber-Valtorta, 2017) for stationary harmonic maps.
- **Other Models:** Our results can be applied to other models, e.g. mean curvature flows.

Remarks on Stratification

- **Top Stratum:** $\Sigma^{n-2}(u, t)$ is the top spatial stratum, implying that tangent measures with spatial $(n - 1)$ -symmetry vanish.
- **Sharpness:** Since $\dim_{\mathcal{H}} \Sigma^k(u, t) \leq k$, the rectifiability and the Minkowski dimension bound are all **sharp**.
- **Via TF:** $S^k(u, t) \subset \Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n - 2]$.
- **Uniqueness of TF:** We do not require the uniqueness of TF.
- **Harmonic Maps:** Assuming $\partial_t u \equiv 0$, the properties degenerate to those in (Naber-Valtorta, 2017) for stationary harmonic maps.
- **Other Models:** Our results can be applied to other models, e.g. mean curvature flows.

Remarks on Stratification

- **Top Stratum:** $\Sigma^{n-2}(u, t)$ is the top spatial stratum, implying that tangent measures with spatial $(n-1)$ -symmetry vanish.
- **Sharpness:** Since $\dim_{\mathcal{H}} \Sigma^k(u, t) \leq k$, the rectifiability and the Minkowski dimension bound are all **sharp**.
- **Via TF:** $S^k(u, t) \subset \Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$.
- **Uniqueness of TF:** We do not require the uniqueness of TF.
- **Harmonic Maps:** Assuming $\partial_t u \equiv 0$, the properties degenerate to those in (Naber-Valtorta, 2017) for stationary harmonic maps.
- **Other Models:** Our results can be applied to other models, e.g. mean curvature flows.

Remarks on Stratification

- **Top Stratum:** $\Sigma^{n-2}(u, t)$ is the top spatial stratum, implying that tangent measures with spatial $(n-1)$ -symmetry vanish.
- **Sharpness:** Since $\dim_{\mathcal{H}} \Sigma^k(u, t) \leq k$, the rectifiability and the Minkowski dimension bound are all **sharp**.
- **Via TF:** $S^k(u, t) \subset \Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$.
- **Uniqueness of TF:** We do not require the uniqueness of TF.
- **Harmonic Maps:** Assuming $\partial_t u \equiv 0$, the properties degenerate to those in (Naber-Valtorta, 2017) for stationary harmonic maps.
- **Other Models:** Our results can be applied to other models, e.g. mean curvature flows.

Remarks on Stratification

- **Top Stratum:** $\Sigma^{n-2}(u, t)$ is the top spatial stratum, implying that tangent measures with spatial $(n-1)$ -symmetry vanish.
- **Sharpness:** Since $\dim_{\mathcal{H}} \Sigma^k(u, t) \leq k$, the rectifiability and the Minkowski dimension bound are all **sharp**.
- **Via TF:** $S^k(u, t) \subset \Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$.
- **Uniqueness of TF:** We do not require the uniqueness of TF.
- **Harmonic Maps:** Assuming $\partial_t u \equiv 0$, the properties degenerate to those in (Naber-Valtorta, 2017) for stationary harmonic maps.
- **Other Models:** Our results can be applied to other models, e.g. mean curvature flows.

Remarks on Stratification

- **Top Stratum:** $\Sigma^{n-2}(u, t)$ is the top spatial stratum, implying that tangent measures with spatial $(n-1)$ -symmetry vanish.
- **Sharpness:** Since $\dim_{\mathcal{H}} \Sigma^k(u, t) \leq k$, the rectifiability and the Minkowski dimension bound are all **sharp**.
- **Via TF:** $S^k(u, t) \subset \Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$.
- **Uniqueness of TF:** We do not require the uniqueness of TF.
- **Harmonic Maps:** Assuming $\partial_t u \equiv 0$, the properties degenerate to those in (Naber-Valtorta, 2017) for stationary harmonic maps.
- **Other Models:** Our results can be applied to other models, e.g. mean curvature flows.

Remarks on Stratification

- **Top Stratum:** $\Sigma^{n-2}(u, t)$ is the top spatial stratum, implying that tangent measures with spatial $(n-1)$ -symmetry vanish.
- **Sharpness:** Since $\dim_{\mathcal{H}} \Sigma^k(u, t) \leq k$, the rectifiability and the Minkowski dimension bound are all **sharp**.
- **Via TF:** $S^k(u, t) \subset \Sigma^k(u, t)$ is k -rectifiable $\forall k \in \mathbb{Z} \cap [1, n-2]$.
- **Uniqueness of TF:** **We do not require the uniqueness of TF.**
- **Harmonic Maps:** Assuming $\partial_t u \equiv 0$, the properties degenerate to those in (Naber-Valtorta, 2017) for stationary harmonic maps.
- **Other Models:** Our results can be applied to other models, e.g. mean curvature flows.

Extension to Ginzburg-Landau Model

Our results also extend to the Chen-Struwe solution. Consider the Ginzburg-Landau model

$$\partial_t u_\varepsilon - \Delta u_\varepsilon = \varepsilon^{-2} f(u_\varepsilon), \quad f(p) = -D_p[\chi(\text{dist}^2(p, \mathcal{N}))],$$

where χ is an appropriate cut-off function.

- **Convergence:** If $\sup_{\varepsilon > 0} \|\nabla u_\varepsilon\|_{L^1(P_1)} \leq \Lambda$, then $\exists \varepsilon_i \rightarrow 0^+$ s.t.

- $u_i := u_{\varepsilon_i} \rightharpoonup u$ weakly in $H^1(P_1)$;
- $\frac{1}{2} |\nabla u_i|^2 dx dt \rightharpoonup^* \mu$ in $\mathcal{M}(P_1)$.

- **Concentration Set:**

$$\Sigma := \bigcap_{r>0} \left\{ X \in P_1 : \liminf_{i \rightarrow +\infty} \left(r^{-n} \int_{P_r(X)} |\nabla u_i|^2 \right) \geq \frac{\varepsilon_0(n, \mathcal{N})}{2} \right\}.$$

- **Analogous Results:** $\Sigma \cap \{t\}$ is $(n-2)$ -rectifiable $\forall t$, with

$$\dim_{\text{Min}}(\Sigma \cap \{t\}) \leq n-2.$$

- **Remark:** (Lin-Wang, 2002) showed rectifiability for a.e. t .

Extension to Ginzburg-Landau Model

Our results also extend to the Chen-Struwe solution. Consider the Ginzburg-Landau model

$$\partial_t u_\varepsilon - \Delta u_\varepsilon = \varepsilon^{-2} f(u_\varepsilon), \quad f(p) = -D_p[\chi(\text{dist}^2(p, \mathcal{N}))],$$

where χ is an appropriate cut-off function.

- **Convergence:** If $\sup_{\varepsilon > 0} \|\nabla u_\varepsilon\|_{L^1(P_1)} \leq \Lambda$, then $\exists \varepsilon_i \rightarrow 0^+$ s.t.

- $u_i := u_{\varepsilon_i} \rightharpoonup u$ weakly in $H^1(P_1)$;
- $\frac{1}{2} |\nabla u_i|^2 dx dt \rightharpoonup^* \mu$ in $\mathcal{M}(P_1)$.

- **Concentration Set:**

$$\Sigma := \bigcap_{r>0} \left\{ X \in P_1 : \liminf_{i \rightarrow +\infty} \left(r^{-n} \int_{P_r(X)} |\nabla u_i|^2 \right) \geq \frac{\varepsilon_0(n, \mathcal{N})}{2} \right\}.$$

- **Analogous Results:** $\Sigma \cap \{t\}$ is $(n-2)$ -rectifiable $\forall t$, with

$$\dim_{\text{Min}}(\Sigma \cap \{t\}) \leq n-2.$$

- **Remark:** (Lin-Wang, 2002) showed rectifiability for a.e. t .

Extension to Ginzburg-Landau Model

Our results also extend to the Chen-Struwe solution. Consider the Ginzburg-Landau model

$$\partial_t u_\varepsilon - \Delta u_\varepsilon = \varepsilon^{-2} f(u_\varepsilon), \quad f(p) = -D_p[\chi(\text{dist}^2(p, \mathcal{N}))],$$

where χ is an appropriate cut-off function.

• **Convergence:** If $\sup_{\varepsilon>0} \|\nabla u_\varepsilon\|_{L^1(P_1)} \leq \Lambda$, then $\exists \varepsilon_i \rightarrow 0^+$ s.t.

- $u_i := u_{\varepsilon_i} \rightharpoonup u$ weakly in $H^1(P_1)$;
- $\frac{1}{2} |\nabla u_i|^2 dx dt \rightharpoonup^* \mu$ in $\mathcal{M}(P_1)$.

• **Concentration Set:**

$$\Sigma := \bigcap_{r>0} \left\{ X \in P_1 : \liminf_{i \rightarrow +\infty} \left(r^{-n} \int_{P_r(X)} |\nabla u_i|^2 \right) \geq \frac{\varepsilon_0(n, \mathcal{N})}{2} \right\}.$$

• **Analogous Results:** $\Sigma \cap \{t\}$ is $(n-2)$ -rectifiable $\forall t$, with

$$\dim_{\text{Min}}(\Sigma \cap \{t\}) \leq n-2.$$

• **Remark:** (Lin-Wang, 2002) showed rectifiability for a.e. t .

Extension to Ginzburg-Landau Model

Our results also extend to the Chen-Struwe solution. Consider the Ginzburg-Landau model

$$\partial_t u_\varepsilon - \Delta u_\varepsilon = \varepsilon^{-2} f(u_\varepsilon), \quad f(p) = -D_p[\chi(\text{dist}^2(p, \mathcal{N}))],$$

where χ is an appropriate cut-off function.

- **Convergence:** If $\sup_{\varepsilon>0} \|\nabla u_\varepsilon\|_{L^1(P_1)} \leq \Lambda$, then $\exists \varepsilon_i \rightarrow 0^+$ s.t.

- $u_i := u_{\varepsilon_i} \rightharpoonup u$ weakly in $H^1(P_1)$;
- $\frac{1}{2} |\nabla u_i|^2 dx dt \rightharpoonup^* \mu$ in $\mathcal{M}(P_1)$.

- **Concentration Set:**

$$\Sigma := \bigcap_{r>0} \left\{ X \in P_1 : \liminf_{i \rightarrow +\infty} \left(r^{-n} \int_{P_r(X)} |\nabla u_i|^2 \right) \geq \frac{\varepsilon_0(n, \mathcal{N})}{2} \right\}.$$

- **Analogous Results:** $\Sigma \cap \{t\}$ is $(n-2)$ -rectifiable $\forall t$, with

$$\dim_{\text{Min}}(\Sigma \cap \{t\}) \leq n-2.$$

- **Remark:** (Lin-Wang, 2002) showed rectifiability for a.e. t .

Extension to Ginzburg-Landau Model

Our results also extend to the Chen-Struwe solution. Consider the Ginzburg-Landau model

$$\partial_t u_\varepsilon - \Delta u_\varepsilon = \varepsilon^{-2} f(u_\varepsilon), \quad f(p) = -D_p[\chi(\text{dist}^2(p, \mathcal{N}))],$$

where χ is an appropriate cut-off function.

• **Convergence:** If $\sup_{\varepsilon > 0} \|\nabla u_\varepsilon\|_{L^1(P_1)} \leq \Lambda$, then $\exists \varepsilon_i \rightarrow 0^+$ s.t.

- $u_i := u_{\varepsilon_i} \rightharpoonup u$ weakly in $H^1(P_1)$;
- $\frac{1}{2} |\nabla u_i|^2 dx dt \rightharpoonup^* \mu$ in $\mathcal{M}(P_1)$.

• **Concentration Set:**

$$\Sigma := \bigcap_{r>0} \left\{ X \in P_1 : \liminf_{i \rightarrow +\infty} \left(r^{-n} \int_{P_r(X)} |\nabla u_i|^2 \right) \geq \frac{\varepsilon_0(n, \mathcal{N})}{2} \right\}.$$

• **Analogous Results:** $\Sigma \cap \{t\}$ is $(n-2)$ -rectifiable $\forall t$, with

$$\dim_{\text{Min}}(\Sigma \cap \{t\}) \leq n-2.$$

• **Remark:** (Lin-Wang, 2002) showed rectifiability for a.e. t .

Extension to Ginzburg-Landau Model

Our results also extend to the Chen-Struwe solution. Consider the Ginzburg-Landau model

$$\partial_t u_\varepsilon - \Delta u_\varepsilon = \varepsilon^{-2} f(u_\varepsilon), \quad f(p) = -D_p[\chi(\text{dist}^2(p, \mathcal{N}))],$$

where χ is an appropriate cut-off function.

• **Convergence:** If $\sup_{\varepsilon>0} \|\nabla u_\varepsilon\|_{L^1(P_1)} \leq \Lambda$, then $\exists \varepsilon_i \rightarrow 0^+$ s.t.

- $u_i := u_{\varepsilon_i} \rightharpoonup u$ weakly in $H^1(P_1)$;
- $\frac{1}{2} |\nabla u_i|^2 dx dt \rightharpoonup^* \mu$ in $\mathcal{M}(P_1)$.

• **Concentration Set:**

$$\Sigma := \bigcap_{r>0} \left\{ X \in P_1 : \liminf_{i \rightarrow +\infty} \left(r^{-n} \int_{P_r(X)} |\nabla u_i|^2 \right) \geq \frac{\varepsilon_0(n, \mathcal{N})}{2} \right\}.$$

• **Analogous Results:** $\Sigma \cap \{t\}$ is $(n-2)$ -rectifiable $\forall t$, with

$$\dim_{\text{Min}}(\Sigma \cap \{t\}) \leq n-2.$$

• **Remark:** (Lin-Wang, 2002) showed rectifiability for a.e. t .

Extension to Ginzburg-Landau Model

Our results also extend to the Chen-Struwe solution. Consider the Ginzburg-Landau model

$$\partial_t u_\varepsilon - \Delta u_\varepsilon = \varepsilon^{-2} f(u_\varepsilon), \quad f(p) = -D_p[\chi(\text{dist}^2(p, \mathcal{N}))],$$

where χ is an appropriate cut-off function.

- **Convergence:** If $\sup_{\varepsilon > 0} \|\nabla u_\varepsilon\|_{L^1(P_1)} \leq \Lambda$, then $\exists \varepsilon_i \rightarrow 0^+$ s.t.

- $u_i := u_{\varepsilon_i} \rightharpoonup u$ weakly in $H^1(P_1)$;
- $\frac{1}{2} |\nabla u_i|^2 dx dt \rightharpoonup^* \mu$ in $\mathcal{M}(P_1)$.

- **Concentration Set:**

$$\Sigma := \bigcap_{r>0} \left\{ X \in P_1 : \liminf_{i \rightarrow +\infty} \left(r^{-n} \int_{P_r(X)} |\nabla u_i|^2 \right) \geq \frac{\varepsilon_0(n, \mathcal{N})}{2} \right\}.$$

- **Analogous Results:** $\Sigma \cap \{t\}$ is $(n-2)$ -rectifiable $\forall t$, with

$$\dim_{\text{Min}}(\Sigma \cap \{t\}) \leq n-2.$$

- **Remark:** (Lin-Wang, 2002) showed rectifiability for a.e. t .

Extension to Ginzburg-Landau Model

Our results also extend to the Chen-Struwe solution. Consider the Ginzburg-Landau model

$$\partial_t u_\varepsilon - \Delta u_\varepsilon = \varepsilon^{-2} f(u_\varepsilon), \quad f(p) = -D_p[\chi(\text{dist}^2(p, \mathcal{N}))],$$

where χ is an appropriate cut-off function.

- **Convergence:** If $\sup_{\varepsilon > 0} \|\nabla u_\varepsilon\|_{L^1(P_1)} \leq \Lambda$, then $\exists \varepsilon_i \rightarrow 0^+$ s.t.

- $u_i := u_{\varepsilon_i} \rightharpoonup u$ weakly in $H^1(P_1)$;
- $\frac{1}{2} |\nabla u_i|^2 dx dt \rightharpoonup^* \mu$ in $\mathcal{M}(P_1)$.

- **Concentration Set:**

$$\Sigma := \bigcap_{r>0} \left\{ X \in P_1 : \liminf_{i \rightarrow +\infty} \left(r^{-n} \int_{P_r(X)} |\nabla u_i|^2 \right) \geq \frac{\varepsilon_0(n, \mathcal{N})}{2} \right\}.$$

- **Analogous Results:** $\Sigma \cap \{t\}$ is $(n-2)$ -rectifiable $\forall t$, with

$$\dim_{\text{Min}}(\Sigma \cap \{t\}) \leq n-2.$$

- **Remark:** (Lin-Wang, 2002) showed rectifiability for a.e. t .

Table of Contents

1 Background and Motivation

- Harmonic Map Flows
- Suitable Solutions
- Stratification Theory

2 Rectifiability and Minkowski Dimension

- Main Results
- Some Remarks

3 Improvements of the Regularity

- Main Results
- Some Remarks

4 Difficulties and Sketch of the Proof

- Difficulties and Strategies
- Sketch of the Proof

Harmonic and Quasi-Harmonic Spheres

$k \in \mathbb{Z} \cap [0, n-1]$.

We recall concepts in (Lin-Wang, 1999).

- **Harmonic k -Sphere:** A non-constant smooth map $\psi : \mathbb{S}^k \rightarrow \mathcal{N}$, critical for

$$\int_{\mathbb{S}^k} |\nabla_{\mathbb{S}^k} \psi|^2 dx.$$

- **Quasi-Harmonic k -Sphere:** A non-constant smooth map $\psi : \mathbb{R}^k \rightarrow \mathcal{N}$, critical for

$$\int_{\mathbb{R}^k} |\nabla \psi|^2 \exp\left(-\frac{|x|^2}{4}\right) dx.$$

- **Note:** If ψ is a quasi-harmonic k -sphere, then

$$u(x, t) = \psi\left(\frac{x}{\sqrt{-t}}\right)$$

solves the (HMF) for $t < 0$.

Harmonic and Quasi-Harmonic Spheres

$k \in \mathbb{Z} \cap [0, n-1]$.

We recall concepts in (Lin-Wang, 1999).

- **Harmonic k -Sphere:** A **non-constant** smooth map $\psi : \mathbb{S}^k \rightarrow \mathcal{N}$, critical for

$$\int_{\mathbb{S}^k} |\nabla_{\mathbb{S}^k} \psi|^2 dx.$$

- **Quasi-Harmonic k -Sphere:** A **non-constant** smooth map $\psi : \mathbb{R}^k \rightarrow \mathcal{N}$, critical for

$$\int_{\mathbb{R}^k} |\nabla \psi|^2 \exp\left(-\frac{|x|^2}{4}\right) dx.$$

- **Note:** If ψ is a quasi-harmonic k -sphere, then

$$u(x, t) = \psi\left(\frac{x}{\sqrt{-t}}\right)$$

solves the (HMF) for $t < 0$.

Harmonic and Quasi-Harmonic Spheres

$k \in \mathbb{Z} \cap [0, n-1]$.

We recall concepts in (Lin-Wang, 1999).

- **Harmonic k -Sphere:** A **non-constant** smooth map $\psi : \mathbb{S}^k \rightarrow \mathcal{N}$, critical for

$$\int_{\mathbb{S}^k} |\nabla_{\mathbb{S}^k} \psi|^2 dx.$$

- **Quasi-Harmonic k -Sphere:** A **non-constant** smooth map $\psi : \mathbb{R}^k \rightarrow \mathcal{N}$, critical for

$$\int_{\mathbb{R}^k} |\nabla \psi|^2 \exp\left(-\frac{|x|^2}{4}\right) dx.$$

- **Note:** If ψ is a quasi-harmonic k -sphere, then

$$u(x, t) = \psi\left(\frac{x}{\sqrt{-t}}\right)$$

solves the (HMF) for $t < 0$.

Harmonic and Quasi-Harmonic Spheres

$k \in \mathbb{Z} \cap [0, n-1]$.

We recall concepts in (Lin-Wang, 1999).

- **Harmonic k -Sphere:** A **non-constant** smooth map $\psi : \mathbb{S}^k \rightarrow \mathcal{N}$, critical for

$$\int_{\mathbb{S}^k} |\nabla_{\mathbb{S}^k} \psi|^2 dx.$$

- **Quasi-Harmonic k -Sphere:** A **non-constant** smooth map $\psi : \mathbb{R}^k \rightarrow \mathcal{N}$, critical for

$$\int_{\mathbb{R}^k} |\nabla \psi|^2 \exp\left(-\frac{|x|^2}{4}\right) dx.$$

- **Note:** If ψ is a quasi-harmonic k -sphere, then

$$u(x, t) = \psi\left(\frac{x}{\sqrt{-t}}\right)$$

solves the (HMF) for $t < 0$.

Main Theorems: Regularity Improvements

Theorem (Fu-W.-Wu-Zhang, arXiv:2504.14880)

$u : P_2 \rightarrow \mathcal{N}$: suitable solution of (HMF).

$\mathcal{N}$: admits *neither harmonic nor quasi-harmonic 2-spheres*.

Then

$$\sup_{t \in (-1,1)} \|\nabla u(\cdot, t)\|_{L^{3,\infty}(B_1)} \leq C(\Lambda, n, \mathcal{N}).$$

The Lorentz space $L^{p,\infty}(E)$ for $p \in [1, \infty)$ and measurable $E \subset \mathbb{R}^n$ is

$$\|f\|_{L^{p,\infty}(E)} := \sup_{t>0} t [\mathcal{L}^n(\{x \in E : |f(x)| > t\})]^{\frac{1}{p}} < +\infty.$$

Since $L^{p,\infty} \subset L^q \forall q \in [1, p)$, we have

$$\sup_{t \in (-1,1)} \|\nabla u(\cdot, t)\|_{L^q(B_1)} \leq C(\Lambda, n, \mathcal{N}, q) \quad \forall q \in [1, 3).$$

Main Theorems: Regularity Improvements

Theorem (Fu-W.-Wu-Zhang, arXiv:2504.14880)

$u : P_2 \rightarrow \mathcal{N}$: suitable solution of (HMF).

$\mathcal{N}$: admits *neither harmonic nor quasi-harmonic 2-spheres*.

Then

$$\sup_{t \in (-1, 1)} \|\nabla u(\cdot, t)\|_{L^{3, \infty}(B_1)} \leq C(\Lambda, n, \mathcal{N}).$$

The Lorentz space $L^{p, \infty}(E)$ for $p \in [1, \infty)$ and measurable $E \subset \mathbb{R}^n$ is

$$\|f\|_{L^{p, \infty}(E)} := \sup_{t > 0} t [\mathcal{L}^n(\{x \in E : |f(x)| > t\})]^{\frac{1}{p}} < +\infty.$$

Since $L^{p, \infty} \subset L^q \forall q \in [1, p)$, we have

$$\sup_{t \in (-1, 1)} \|\nabla u(\cdot, t)\|_{L^q(B_1)} \leq C(\Lambda, n, \mathcal{N}, q) \quad \forall q \in [1, 3).$$

Remarks on the Regularity Improvement

- **Role of Absence:**

- *Harmonic 2-Spheres:* Their absence ensures vanishing defect measures.
- *Quasi-Harmonic 2-Spheres:* Prevents existence of non-trivial spatially $(n-2)$ -symmetric tangent flows.

- **Regularity Implications:**

- *Static Case:* (Naber-Valtorta, 2017) proved regularity for Dirichlet energy minimizers.
- *Dynamic Case:* Our results leverage $\mathcal{N}$ properties for sharp regularity bounds.

- **Note:** If we assume the absence of quasi-harmonic $\mathbb{S}^k$ -spheres, then the regularity can be further improved.

- **Summary:** The absence of 2-spheres simplifies tangent flow analysis, yielding improved regularity for harmonic map flows.

Remarks on the Regularity Improvement

- **Role of Absence:**

- *Harmonic 2-Spheres:* Their absence ensures vanishing defect measures.
- *Quasi-Harmonic 2-Spheres:* Prevents existence of non-trivial spatially $(n-2)$ -symmetric tangent flows.

- **Regularity Implications:**

- *Static Case:* (Naber-Valtorta, 2017) proved regularity for Dirichlet energy minimizers.
- *Dynamic Case:* Our results leverage $\mathcal{N}$ properties for sharp regularity bounds.
- **Note:** If we assume the absence of quasi-harmonic $\mathbb{S}^k$ -spheres, then the regularity can be further improved.
- **Summary:** The absence of 2-spheres simplifies tangent flow analysis, yielding improved regularity for harmonic map flows.

Remarks on the Regularity Improvement

- **Role of Absence:**

- *Harmonic 2-Spheres:* Their absence ensures **vanishing defect measures**.
- *Quasi-Harmonic 2-Spheres:* Prevents existence of **non-trivial spatially $(n-2)$ -symmetric tangent flows**.

- **Regularity Implications:**

- *Static Case:* (Naber-Valtorta, 2017) proved regularity for Dirichlet energy minimizers.
- *Dynamic Case:* Our results leverage $\mathcal{N}$ properties for sharp regularity bounds.
- **Note:** If we assume the **absence of quasi-harmonic $\mathbb{S}^k$ -spheres**, then the regularity can be further improved.
- **Summary:** The absence of 2-spheres simplifies tangent flow analysis, yielding improved regularity for harmonic map flows.

Remarks on the Regularity Improvement

- **Role of Absence:**

- *Harmonic 2-Spheres*: Their absence ensures **vanishing defect measures**.
- *Quasi-Harmonic 2-Spheres*: Prevents existence of **non-trivial spatially $(n - 2)$ -symmetric tangent** flows.

- **Regularity Implications:**

- *Static Case*: (Naber-Valtorta, 2017) proved regularity for Dirichlet energy minimizers.
- *Dynamic Case*: Our results leverage $\mathcal{N}$ properties for sharp regularity bounds.
- **Note**: If we assume the **absence of quasi-harmonic $\mathbb{S}^k$ -spheres**, then the regularity can be further improved.
- **Summary**: The absence of 2-spheres simplifies tangent flow analysis, yielding improved regularity for harmonic map flows.

Remarks on the Regularity Improvement

- **Role of Absence:**

- *Harmonic 2-Spheres*: Their absence ensures **vanishing defect measures**.
- *Quasi-Harmonic 2-Spheres*: Prevents existence of **non-trivial spatially $(n - 2)$ -symmetric tangent** flows.

- **Regularity Implications:**

- *Static Case*: (Naber-Valtorta, 2017) proved regularity for Dirichlet energy minimizers.
- *Dynamic Case*: Our results leverage $\mathcal{N}$ properties for sharp regularity bounds.
- **Note**: If we assume the **absence of quasi-harmonic $\mathbb{S}^k$ -spheres**, then the regularity can be further improved.
- **Summary**: The absence of 2-spheres simplifies tangent flow analysis, yielding improved regularity for harmonic map flows.

Remarks on the Regularity Improvement

- **Role of Absence:**

- *Harmonic 2-Spheres*: Their absence ensures **vanishing defect measures**.
- *Quasi-Harmonic 2-Spheres*: Prevents existence of **non-trivial spatially $(n - 2)$ -symmetric tangent** flows.

- **Regularity Implications:**

- *Static Case*: (Naber-Valtorta, 2017) proved regularity for Dirichlet energy minimizers.
- *Dynamic Case*: Our results leverage $\mathcal{N}$ properties for sharp regularity bounds.
- **Note**: If we assume the **absence of quasi-harmonic $\mathbb{S}^k$ -spheres**, then the regularity can be further improved.
- **Summary**: The absence of 2-spheres simplifies tangent flow analysis, yielding improved regularity for harmonic map flows.

Remarks on the Regularity Improvement

- **Role of Absence:**

- *Harmonic 2-Spheres*: Their absence ensures **vanishing defect measures**.
- *Quasi-Harmonic 2-Spheres*: Prevents existence of **non-trivial spatially $(n - 2)$ -symmetric tangent** flows.

- **Regularity Implications:**

- *Static Case*: (Naber-Valtorta, 2017) proved regularity for Dirichlet energy minimizers.
- *Dynamic Case*: Our results leverage $\mathcal{N}$ properties for sharp regularity bounds.
- **Note**: If we assume the **absence of quasi-harmonic $\mathbb{S}^k$ -spheres**, then the regularity can be further improved.
- **Summary**: The absence of 2-spheres simplifies tangent flow analysis, yielding improved regularity for harmonic map flows.

Remarks on the Regularity Improvement

- **Role of Absence:**

- *Harmonic 2-Spheres*: Their absence ensures **vanishing defect measures**.
- *Quasi-Harmonic 2-Spheres*: Prevents existence of **non-trivial spatially $(n - 2)$ -symmetric tangent** flows.

- **Regularity Implications:**

- *Static Case*: (Naber-Valtorta, 2017) proved regularity for Dirichlet energy minimizers.
- *Dynamic Case*: Our results leverage $\mathcal{N}$ properties for sharp regularity bounds.

- **Note**: If we assume the **absence of quasi-harmonic $\mathbb{S}^k$ -spheres**, then the regularity can be further improved.

- **Summary**: The absence of 2-spheres simplifies tangent flow analysis, yielding improved regularity for harmonic map flows.

Remarks on the Regularity Improvement

- **Role of Absence:**

- *Harmonic 2-Spheres*: Their absence ensures **vanishing defect measures**.
- *Quasi-Harmonic 2-Spheres*: Prevents existence of **non-trivial spatially $(n-2)$ -symmetric tangent** flows.

- **Regularity Implications:**

- *Static Case*: (Naber-Valtorta, 2017) proved regularity for Dirichlet energy minimizers.
- *Dynamic Case*: Our results leverage $\mathcal{N}$ properties for sharp regularity bounds.

- **Note**: If we assume the **absence of quasi-harmonic $\mathbb{S}^k$ -spheres**, then the regularity can be further improved.

- **Summary**: The absence of 2-spheres simplifies tangent flow analysis, yielding improved regularity for harmonic map flows.

Table of Contents

1 Background and Motivation

- Harmonic Map Flows
- Suitable Solutions
- Stratification Theory

2 Rectifiability and Minkowski Dimension

- Main Results
- Some Remarks

3 Improvements of the Regularity

- Main Results
- Some Remarks

4 Difficulties and Sketch of the Proof

- Difficulties and Strategies
- Sketch of the Proof

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification

- (Cheeger-Naber, 2013): Gromov-Hausdorff limit.

- (Cheeger-Naber, 2013): Harmonic maps and minimal currents.

- Reifenberg-type results (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- Nesting property in the monotonicity formula.

- Unique continuation for weak solutions and measures.

- Scaling invariance in localized density.

- Non-uniqueness of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).

- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.

- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification
 - (Cheeger-Naber, 2013): Gromov-Hausdorff limit.
 - (Cheeger-Naber, 2013): Harmonic maps and minimal currents.
 - **Reifenberg-type results** (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- **Nesting property** in the monotonicity formula.
- **Unique continuation** for weak solutions and measures.
- **Scaling invariance** in localized density.
- **Non-uniqueness** of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).

- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.

- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification

- (Cheeger-Naber, 2013): Gromov-Hausdorff limit.
 - (Cheeger-Naber, 2013): Harmonic maps and minimal currents.
 - Reifenberg-type results (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- Nesting property in the monotonicity formula.
 - Unique continuation for weak solutions and measures.
 - Scaling invariance in localized density.
 - Non-uniqueness of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).

- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.

- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification

- (Cheeger-Naber, 2013): Gromov-Hausdorff limit.

- (Cheeger-Naber, 2013): Harmonic maps and minimal currents.

- Reifenberg-type results (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- Nesting property in the monotonicity formula.

- Unique continuation for weak solutions and measures.

- Scaling invariance in localized density.

- Non-uniqueness of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).

- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.

- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification

- (Cheeger-Naber, 2013): Gromov-Hausdorff limit.

- (Cheeger-Naber, 2013): Harmonic maps and minimal currents.

- Reifenberg-type results (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- Nesting property in the monotonicity formula.

- Unique continuation for weak solutions and measures.

- Scaling invariance in localized density.

- Non-uniqueness of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).

- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.

- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification
 - (Cheeger-Naber, 2013): Gromov-Hausdorff limit.
 - (Cheeger-Naber, 2013): Harmonic maps and minimal currents.
- **Reifenberg-type results** (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- **Nesting property** in the monotonicity formula.
- **Unique continuation** for weak solutions and measures.
- **Scaling invariance** in localized density.
- **Non-uniqueness** of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).

- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.

- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification
 - (Cheeger-Naber, 2013): Gromov-Hausdorff limit.
 - (Cheeger-Naber, 2013): Harmonic maps and minimal currents.
- **Reifenberg-type results** (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- **Nesting property** in the monotonicity formula.
- **Unique continuation** for weak solutions and measures.
- **Scaling invariance** in localized density.
- **Non-uniqueness** of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).

- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.

- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification
 - (Cheeger-Naber, 2013): Gromov-Hausdorff limit.
 - (Cheeger-Naber, 2013): Harmonic maps and minimal currents.
- **Reifenberg-type results** (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- **Nesting property** in the monotonicity formula.
- **Unique continuation** for weak solutions and measures.
- **Scaling invariance** in localized density.
- **Non-uniqueness** of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).

- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.

- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification
 - (Cheeger-Naber, 2013): Gromov-Hausdorff limit.
 - (Cheeger-Naber, 2013): Harmonic maps and minimal currents.
- **Reifenberg-type results** (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- **Nesting property** in the monotonicity formula.
- **Unique continuation** for weak solutions and measures.
- **Scaling invariance** in localized density.
- **Non-uniqueness** of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).
- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.
- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification
 - (Cheeger-Naber, 2013): Gromov-Hausdorff limit.
 - (Cheeger-Naber, 2013): Harmonic maps and minimal currents.
- **Reifenberg-type results** (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- **Nesting property** in the monotonicity formula.
- **Unique continuation** for weak solutions and measures.
- **Scaling invariance** in localized density.
- **Non-uniqueness** of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).
- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.
- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification
 - (Cheeger-Naber, 2013): Gromov-Hausdorff limit.
 - (Cheeger-Naber, 2013): Harmonic maps and minimal currents.
- **Reifenberg-type results** (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- **Nesting property** in the monotonicity formula.
- **Unique continuation** for weak solutions and measures.
- **Scaling invariance** in localized density.
- **Non-uniqueness** of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).

- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.

- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification
 - (Cheeger-Naber, 2013): Gromov-Hausdorff limit.
 - (Cheeger-Naber, 2013): Harmonic maps and minimal currents.
- **Reifenberg-type results** (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- **Nesting property** in the monotonicity formula.
- **Unique continuation** for weak solutions and measures.
- **Scaling invariance** in localized density.
- **Non-uniqueness** of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).

- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.

- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Challenges in Harmonic Map Flow Analysis

- **Key Tools:**

- Quantitative stratification
 - (Cheeger-Naber, 2013): Gromov-Hausdorff limit.
 - (Cheeger-Naber, 2013): Harmonic maps and minimal currents.
- **Reifenberg-type results** (Naber-Valtorta, 2017): Harmonic maps.

- **Dynamic Nature:** Unlike some classical models, flows lack:

- **Nesting property** in the monotonicity formula.
- **Unique continuation** for weak solutions and measures.
- **Scaling invariance** in localized density.
- **Non-uniqueness** of TFs (distinct from (Colding-Minicozzi, 2016) and (Huang-Jiang, arXiv:2406.05877)).

- **Stratification Issues:** Classical methods fail due to unrestricted scales and the compactness is not satisfactory.

- **Remark:** For $n = 3$, one can apply results for approximate harmonic maps (Naber-Valtorta, 2018).

Overcoming Challenges

- **Nonlinear Free Boundary Techniques:** Adapt arguments from further results (Edelen-Engelstein, 2019) to handle the missing **nesting property** and **unique continuation**.
- **Parabolic L^2 -best Estimates:** We establish the L^2 -best estimates in the parabolic form to connect Reifenberg-type results and harmonic map flows.
- **Scale-Connected Strata:** Establish inclusion results for quantitative strata across scales, improved forms of classical ones in (Cheeger-Naber, 2013).
- **Blow-Up Analysis:** Transform local problems into global ones, suiting the dynamic nature of flows, addressing lack of the scaling invariance.
- **Remark:** For some static models like semilinear elliptic equations (Fu-W.-Zhang, arXiv:2408.06726) and (W.-Zhang, arXiv:2411.16048), we mainly apply arguments in (Hirsch-Stuvard-Valtorta, 2019) to deal with some similar difficulties.

Overcoming Challenges

- **Nonlinear Free Boundary Techniques:** Adapt arguments from further results (Edelen-Engelstein, 2019) to handle the missing **nesting property** and **unique continuation**.
- **Parabolic L^2 -best Estimates:** We establish the **L^2 -best estimates in the parabolic form** to connect Reifenberg-type results and harmonic map flows.
- **Scale-Connected Strata:** Establish inclusion results for quantitative strata across scales, improved forms of classical ones in (Cheeger-Naber, 2013).
- **Blow-Up Analysis:** Transform local problems into global ones, suiting the dynamic nature of flows, addressing lack of the scaling invariance.
- **Remark:** For some static models like semilinear elliptic equations (Fu-W.-Zhang, arXiv:2408.06726) and (W.-Zhang, arXiv:2411.16048), we mainly apply arguments in (Hirsch-Stuvard-Valtorta, 2019) to deal with some similar difficulties.

Overcoming Challenges

- **Nonlinear Free Boundary Techniques:** Adapt arguments from further results (Edelen-Engelstein, 2019) to handle the missing **nesting property** and **unique continuation**.
- **Parabolic L^2 -best Estimates:** We establish the **L^2 -best estimates in the parabolic form** to connect Reifenberg-type results and harmonic map flows.
- **Scale-Connected Strata:** Establish inclusion results for quantitative strata **across scales**, improved forms of classical ones in (Cheeger-Naber, 2013).
- **Blow-Up Analysis:** Transform local problems into global ones, suiting the **dynamic nature of flows**, **addressing lack of the scaling invariance**.
- **Remark:** For some static models like semilinear elliptic equations (Fu-W.-Zhang, arXiv:2408.06726) and (W.-Zhang, arXiv:2411.16048), we mainly apply arguments in (Hirsch-Stuvard-Valtorta, 2019) to deal with some similar difficulties.

Overcoming Challenges

- **Nonlinear Free Boundary Techniques:** Adapt arguments from further results (Edelen-Engelstein, 2019) to handle the missing **nesting property** and **unique continuation**.
- **Parabolic L^2 -best Estimates:** We establish the **L^2 -best estimates in the parabolic form** to connect Reifenberg-type results and harmonic map flows.
- **Scale-Connected Strata:** Establish inclusion results for quantitative strata **across scales**, improved forms of classical ones in (Cheeger-Naber, 2013).
- **Blow-Up Analysis:** Transform local problems into global ones, suiting the **dynamic nature** of flows, **addressing lack of the scaling invariance**.
- **Remark:** For some static models like semilinear elliptic equations (Fu-W.-Zhang, arXiv:2408.06726) and (W.-Zhang, arXiv:2411.16048), we mainly apply arguments in (Hirsch-Stuvard-Valtorta, 2019) to deal with some similar difficulties.

Overcoming Challenges

- **Nonlinear Free Boundary Techniques:** Adapt arguments from further results (Edelen-Engelstein, 2019) to handle the missing [nesting property](#) and [unique continuation](#).
- **Parabolic L^2 -best Estimates:** We establish the [\$L^2\$ -best estimates in the parabolic form](#) to connect Reifenberg-type results and harmonic map flows.
- **Scale-Connected Strata:** Establish inclusion results for quantitative strata [across scales](#), improved forms of classical ones in (Cheeger-Naber, 2013).
- **Blow-Up Analysis:** Transform local problems into global ones, suiting the [dynamic nature](#) of flows, [addressing lack of the scaling invariance](#).
- **Remark:** For some static models like semilinear elliptic equations (Fu-W.-Zhang, arXiv:2408.06726) and (W.-Zhang, arXiv:2411.16048), we mainly apply arguments in (Hirsch-Stuvard-Valtorta, 2019) to deal with some similar difficulties.

Quantitative Stratification: Definitions

- **Quantitative Spatial Symmetry:** A map $u \in L^2(P_R, \mathcal{N})$ is spatially (k, ε) -symmetric in $B_r(x)$ at t if there is a spatially k -symmetric measure $\mu \in \mathcal{M}(\mathbb{R}^n \times \mathbb{R})$ s.t.

$$\mathbb{M} \left(\frac{1}{2} |\nabla(T_{X,r}u)|^2 dx dt, \mu \right) < \varepsilon,$$

where $\mathbb{M}(\cdot, \cdot)$ denotes the metric of Radon measures.

- **Quantitative Spatial Stratification:** $0 < r < R \leq 1$, $k \in \mathbb{Z} \cap [0, n-1]$, and $u : P_4 \rightarrow \mathcal{N}$ is a suitable solution. Let

$$\Sigma_{\varepsilon; r, R}^k(u, t) := \{x \in B_2 : u \text{ is not spatially } (k, \varepsilon)\text{-sym in } B_s(x) \text{ at } t, \forall s \in [r, R]\}$$

Define

$$\Sigma_{\varepsilon; 0, R}^k(u, t) := \bigcap_{0 < r < R} \Sigma_{\varepsilon; r, R}^k(u, t).$$

Quantitative Stratification: Definitions

- **Quantitative Spatial Symmetry:** A map $u \in L^2(P_R, \mathcal{N})$ is spatially (k, ε) -symmetric in $B_r(x)$ at t if there is a spatially k -symmetric measure $\mu \in \mathcal{M}(\mathbb{R}^n \times \mathbb{R})$ s.t.

$$\mathbb{M} \left(\frac{1}{2} |\nabla(T_{X,r}u)|^2 dx dt, \mu \right) < \varepsilon,$$

where $\mathbb{M}(\cdot, \cdot)$ denotes the metric of Radon measures.

- **Quantitative Spatial Stratification:** $0 < r < R \leq 1$, $k \in \mathbb{Z} \cap [0, n-1]$, and $u : P_4 \rightarrow \mathcal{N}$ is a suitable solution. Let

$$\Sigma_{\varepsilon; r, R}^k(u, t) := \{x \in B_2 : u \text{ is not spatially } (k, \varepsilon)\text{-sym in } B_s(x) \text{ at } t, \forall s \in [r, R]\}$$

Define

$$\Sigma_{\varepsilon; 0, R}^k(u, t) := \bigcap_{0 < r < R} \Sigma_{\varepsilon; r, R}^k(u, t).$$

Quantitative Stratification: Main Results

Through compactness arguments,

$$\Sigma^k(u, t) \cap B_2 = \bigcup_{\varepsilon > 0} \Sigma_{\varepsilon; 0, R}^k(u, t) = \bigcup_{\varepsilon > 0} \bigcap_{0 < r < R} \Sigma_{\varepsilon; r, R}^k(u, t) \quad \forall R \in (0, 1).$$

Theorem

$\Lambda > 0$, $k \in \mathbb{Z} \cap [1, n-1]$, $R \in (0, 1]$, and $t \in (-4, 4)$.

$u : P_4 \rightarrow \mathcal{N}$: suitable solution of (HMF) with $\|\nabla u\|_{L^2(P_4)} \leq \Lambda$.

- $\forall \varepsilon > 0, r \in (0, R)$,

$$\mathcal{L}^n(B_r(\Sigma_{\varepsilon; r, R}^k(u, t) \cap B_1)) \leq C(\varepsilon, \Lambda, n, \mathcal{N}, R) r^{n-k},$$

$$\mathcal{L}^n(B_r(\Sigma_{\varepsilon; 0, R}^k(u, t) \cap B_1)) \leq C(\varepsilon, \Lambda, n, \mathcal{N}, R) r^{n-k}.$$

- $\forall x \in B_1, \rho \in (0, 1)$,

$$\mathcal{H}^k(\Sigma_{\varepsilon; 0, R}^k(u, t) \cap B_\rho(x)) \leq (\varepsilon, \Lambda, n, \mathcal{N}, R) \rho^k.$$

Quantitative Stratification: Main Results

Through compactness arguments,

$$\Sigma^k(u, t) \cap B_2 = \bigcup_{\varepsilon > 0} \Sigma_{\varepsilon; 0, R}^k(u, t) = \bigcup_{\varepsilon > 0} \bigcap_{0 < r < R} \Sigma_{\varepsilon; r, R}^k(u, t) \quad \forall R \in (0, 1).$$

Theorem

$\Lambda > 0$, $k \in \mathbb{Z} \cap [1, n-1]$, $R \in (0, 1]$, and $t \in (-4, 4)$.

$u : P_4 \rightarrow \mathcal{N}$: suitable solution of (HMF) with $\|\nabla u\|_{L^2(P_4)} \leq \Lambda$.

- $\forall \varepsilon > 0, r \in (0, R)$,

$$\mathcal{L}^n(B_r(\Sigma_{\varepsilon; r, R}^k(u, t) \cap B_1)) \leq C(\varepsilon, \Lambda, n, \mathcal{N}, R) r^{n-k},$$

$$\mathcal{L}^n(B_r(\Sigma_{\varepsilon; 0, R}^k(u, t) \cap B_1)) \leq C(\varepsilon, \Lambda, n, \mathcal{N}, R) r^{n-k}.$$

- $\forall x \in B_1, \rho \in (0, 1)$,

$$\mathcal{H}^k(\Sigma_{\varepsilon; 0, R}^k(u, t) \cap B_\rho(x)) \leq (\varepsilon, \Lambda, n, \mathcal{N}, R) \rho^k.$$

Parabolic L^2 -Best Estimates

Definition (k -Dimensional Displacement)

$U \subset \mathbb{R}^n$: bounded open set. $\mu \in \mathcal{M}(U)$ and $x_0 \in U$.

Define

$$D_\mu^k(x_0, r) := \min_{L: k\text{-dim affine subspaces}} \left(r^{-k-2} \int_{B_r(x_0)} \text{dist}^2(y, L) d\mu(y) \right).$$

Proposition

$\Lambda > 0$. $u : P_4 \rightarrow \mathcal{N}$: suitable solution of (HMF) with $\|\nabla u\|_{L^2(P_4)} \leq \Lambda$.

$\forall \varepsilon > 0$, $\exists \delta = \delta(\varepsilon, \Lambda, n, \mathcal{N}) > 0$ s.t. the following properties hold.

If u is spatially $(0, \delta)$ -sym but not spatially $(k+1, \varepsilon)$ -sym in $B_{2r}(x_0)$ at t_0 , then

$$D_\mu^k(x_0, r) \leq \frac{C}{r^k} \int_{B_r(x_0)} \left[\mathcal{W}\left(u, (y, t_0), 2r, \frac{r}{2}\right) + r \right] d\mu(y),$$

where $\mathcal{W}(u, (y, t_0), 2r, \frac{r}{2})$ represents the difference of densities in scales $2r$ and $\frac{r}{2}$.

Parabolic L^2 -Best Estimates

Definition (k -Dimensional Displacement)

$U \subset \mathbb{R}^n$: bounded open set. $\mu \in \mathcal{M}(U)$ and $x_0 \in U$.

Define

$$D_\mu^k(x_0, r) := \min_{L: k\text{-dim affine subspaces}} \left(r^{-k-2} \int_{B_r(x_0)} \text{dist}^2(y, L) d\mu(y) \right).$$

Proposition

$\Lambda > 0$. $u : P_4 \rightarrow \mathcal{N}$: suitable solution of (HMF) with $\|\nabla u\|_{L^2(P_4)} \leq \Lambda$.

$\forall \varepsilon > 0$, $\exists \delta = \delta(\varepsilon, \Lambda, n, \mathcal{N}) > 0$ s.t. the following properties hold.

If u is spatially $(0, \delta)$ -sym but not spatially $(k+1, \varepsilon)$ -sym in $B_{2r}(x_0)$ at t_0 , then

$$D_\mu^k(x_0, r) \leq \frac{C}{r^k} \int_{B_r(x_0)} \left[\mathcal{W}\left(u, (y, t_0), 2r, \frac{r}{2}\right) + r \right] d\mu(y),$$

where $\mathcal{W}(u, (y, t_0), 2r, \frac{r}{2})$ represents the difference of densities in scales $2r$ and $\frac{r}{2}$.

Thank you for listening!